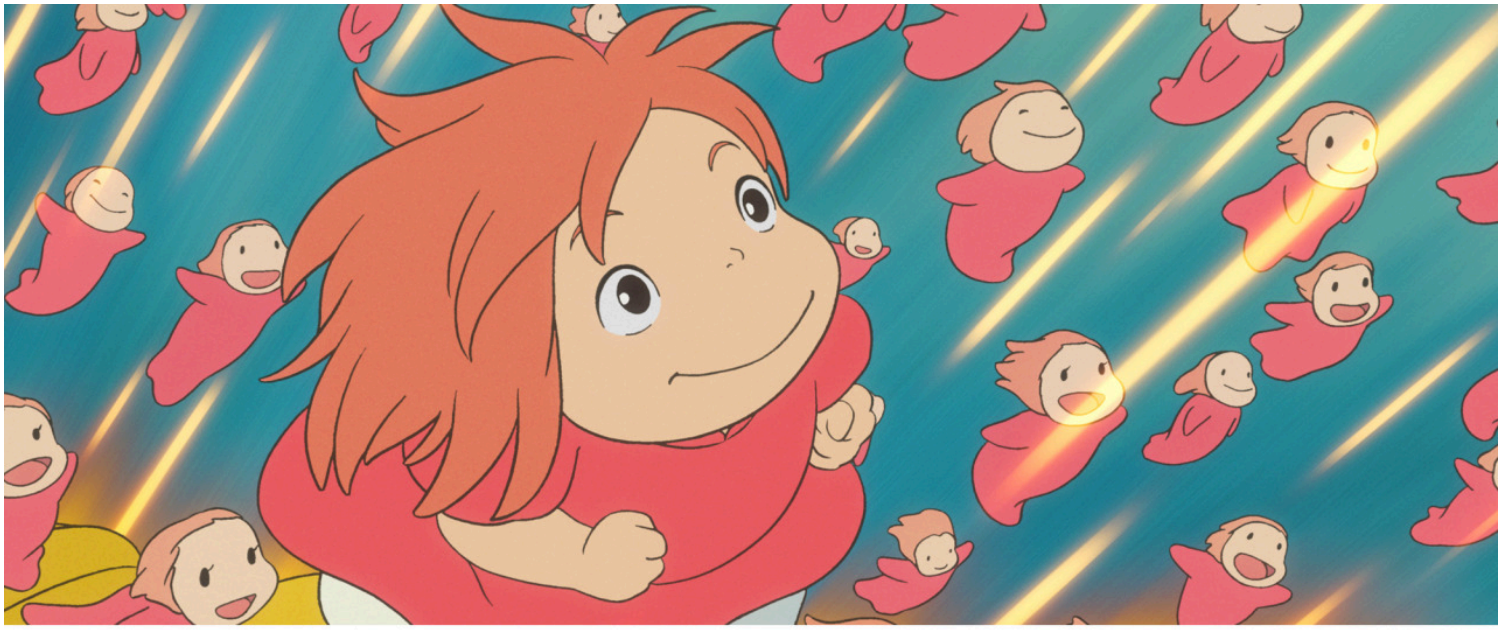
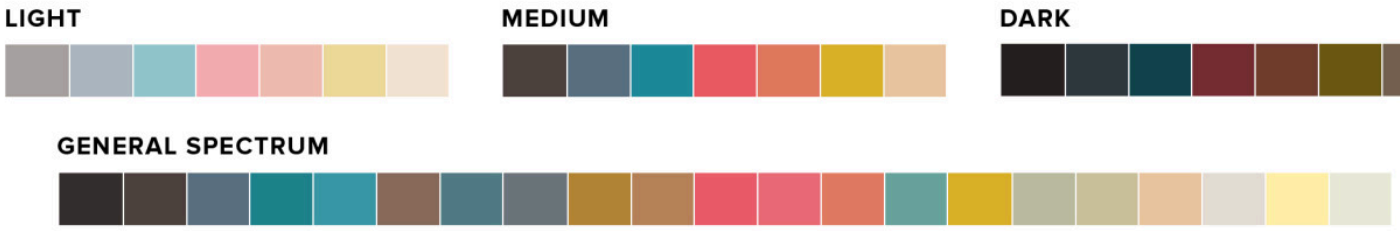
Atomic Physics

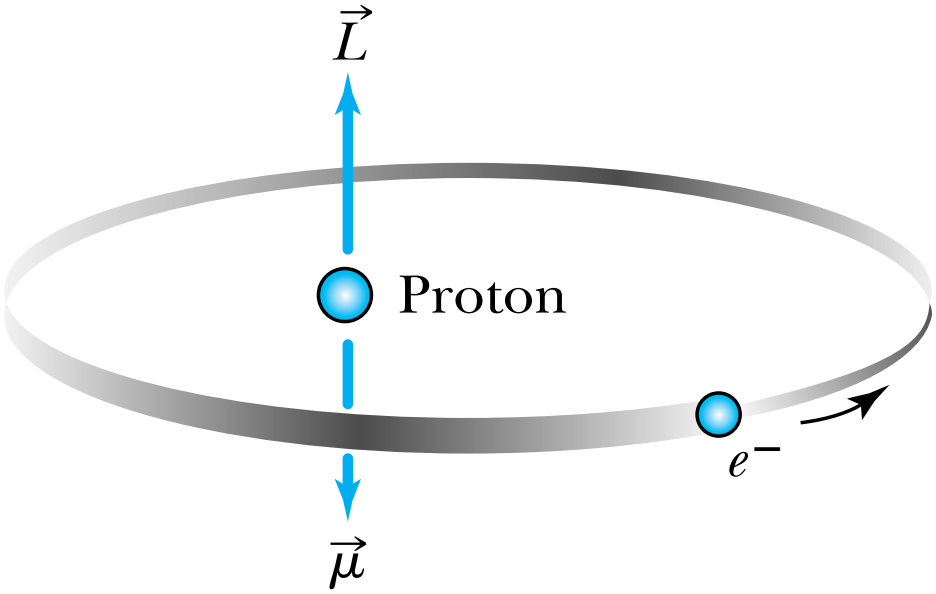
Chapter 4 Fine structures of Atoms





As a rough model, think of an electron circulating around the nucleus as a circular current loop. The current loop has a magnetic moment

$$\mu = IA = \frac{q}{T}A$$
$$= \frac{-e\pi r^2}{2\pi r/v} = \frac{-erv}{2}$$
$$= -\frac{e}{2m}L$$



where L=mvr is the magnitude of the orbital angular momentum.

Magnetic moment of electron

Both the magnetic moment and angular momentum are vectors so that

$$\vec{\mu} = -\frac{e}{2m}\vec{L}$$

Gyromagnetic ratio

$$\gamma = \frac{e}{2m}$$

In classical electromagnetism, if a magnetic dipole having a magnetic moment, is placed in an external magnetic field, the dipole will experience a torque

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

which aligns the dipole with the magnetic field.

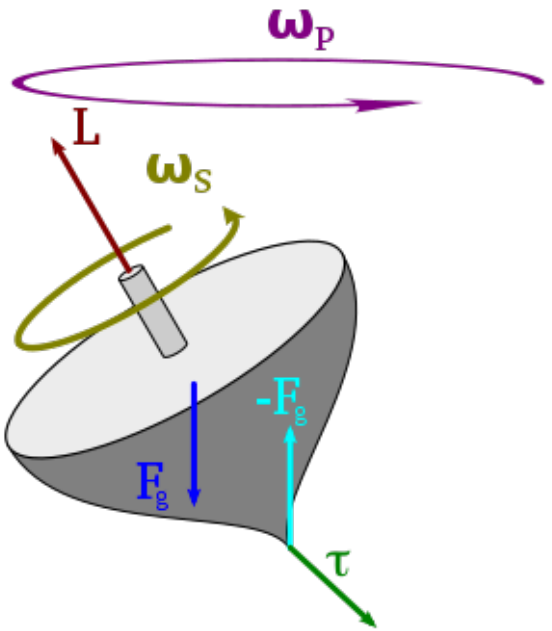
Magnetic moment of electron

The dipole also has a potential energy in the field given by

$$V_B = -\vec{\mu} \cdot \vec{B}$$

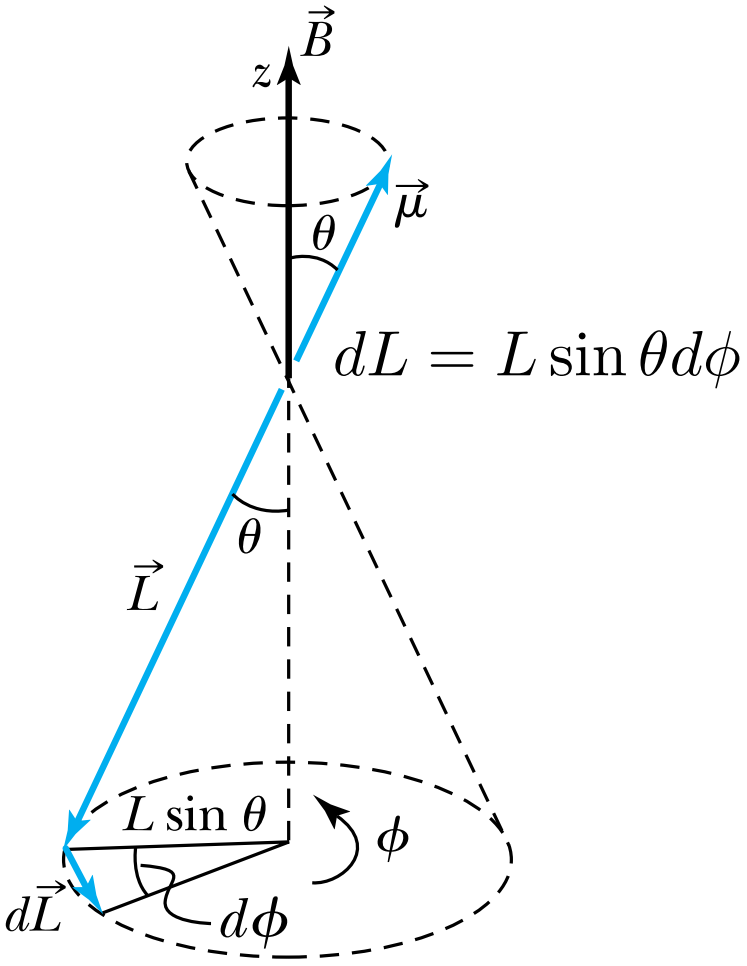
According to classical physics, if the system can change its potential energy, the magnetic moment will align itself with the external magnetic field to minimize energy.

Precession in gravitational field



Magnetic moment of electron

An atom with magnetic moment in an external magnetic field has Larmor precession.



Precessional frequency

$$\omega_L = \frac{d\phi}{dt} = \frac{1}{L\sin\theta}\frac{dL}{dt}$$

Newton's second law in angular form

$$\vec{\tau} = \frac{d\vec{L}}{dt}$$

So

$$\omega_L = \left(\frac{e}{2m\mu\sin\theta}\right)\mu B\sin\theta = \frac{eB}{2m}$$

Orbit angular momentum operator

$$\hat{L} = \hat{r} \times \hat{p}$$

The components of orbit angular moment

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y = -i\hbar\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)$$

$$\hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z = -i\hbar\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right)$$

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x = -i\hbar\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right)$$

The square of orbit angular momentum operator

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$$

Magnetic moment of electron

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Eigenstate and eigenvalue

$$\hat{Q}f = \lambda f$$

f is the eigenstate and λ is the corresponding eigenvalue.

The eigenstate and eigenvalue of orbit angular moment

$$\hat{L}^2 Y_{lm} = L^2 Y_{lm} = l(l+1)\hbar^2 Y_{lm}$$

$$\hat{L}_z Y_{lm} = m_l \hbar Y_{lm}$$

l is the quantum number of orbit angular momentum

m_l is the magnetic quantum number

Magnetic moment of electron

The expectation value of the square of the angular momentum L is therefore

$$\left\langle \boldsymbol{L}^2 \right\rangle = \int \psi^* \hat{L}^2 \psi \, \mathrm{d}\tau = l(l+1)\hbar^2$$

For the absolute value of the angular momentum we obtain

$$|\boldsymbol{L}| = \sqrt{l(l+1)}\hbar.$$

The expectation value of L_z is

$$\langle L_z \rangle = m\hbar.$$

Magnetic moment of electron

If the operators L_x or L_y are applied to the wave function, one can prove that ψ is not an eigenfunction of L_x, i.e.,

$$\hat{L}_x \psi \neq m_x \psi .$$

The same is true for L_y. For their square summation operator,

$$\left(\hat{L}_x^2 + \hat{L}_y^2\right)\psi = \left[l(l+1) - m^2\right]\hbar^2 \psi .$$

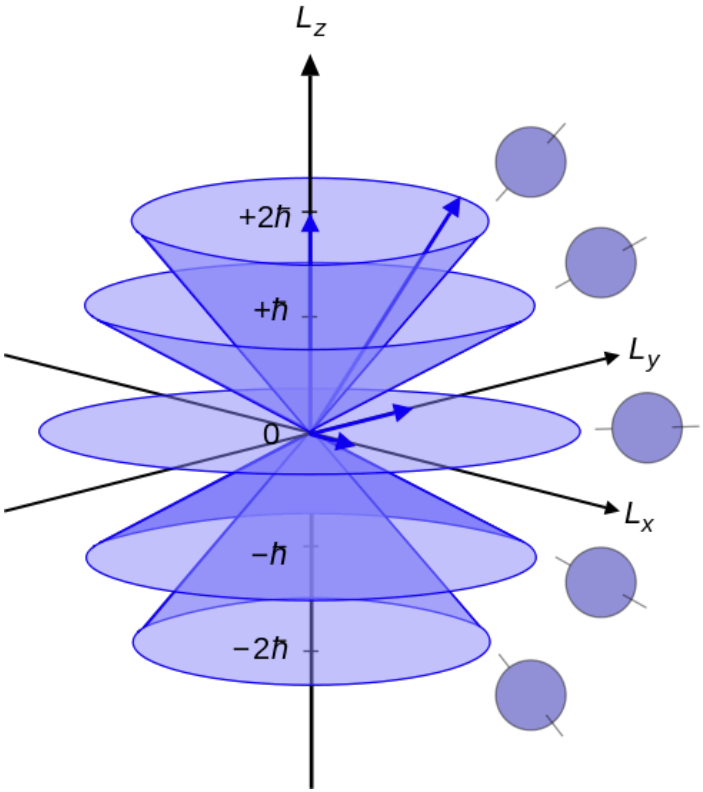
l	m	Name	Degree of degeneracy
0	0	s	1
1	$-1, 0, +1$	p	3
2	-2 to $+2$	d	5
3	-3 to $+3$	f	7
4	-4 to $+4$	g	9
5	-5 to $+5$	h	11

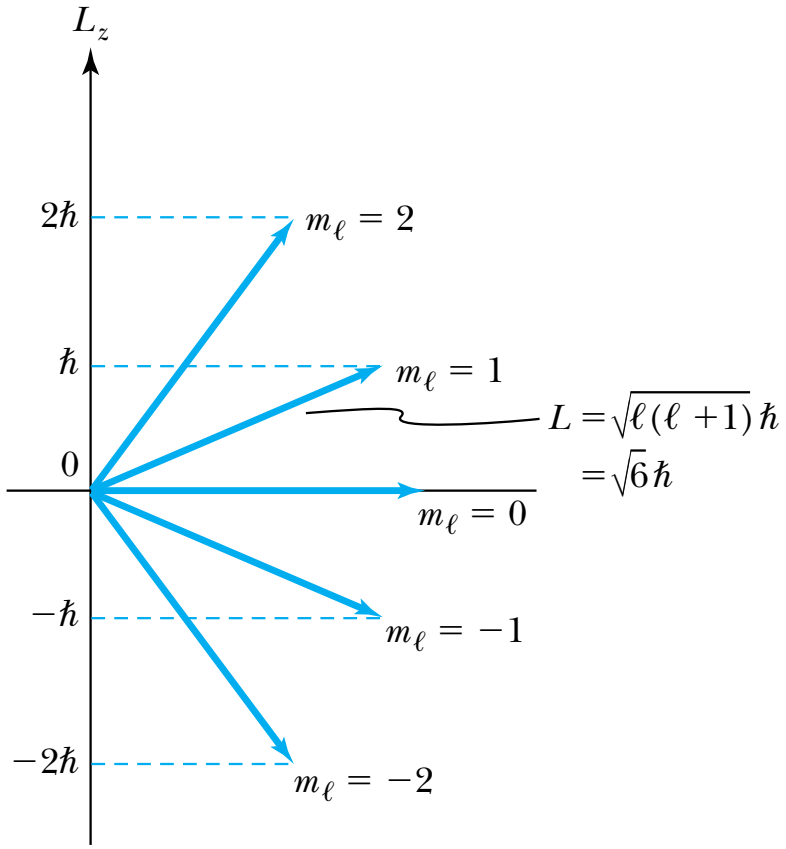
Magnetic moment of electron

$$n = 1, 2, 3, 4, \ldots \qquad \text{Integer}$$

$$l = 0, 1, 2, 3, \ldots, n-1 \qquad \text{Integer}$$

$$m_l = -l, -l+1, \ldots, 0, 1, \ldots, l-1, l \qquad \text{Integer}$$





What are the possible quantum numbers for the state

n=4 in atomic hydrogen?

n	ℓ	m_ℓ
4	0	0
4	1	$-1, 0, 1$
4	2	$-2, -1, 0, 1, 2$
4	3	$-3, -2, -1, 0, 1, 2, 3$

Magnetic moment of electron

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Magnetic moment operator

$$\hat{\vec{\mu}} = -\frac{e}{2m}\hat{\vec{L}}$$

$$\hat{\mu}_z = -\frac{e}{2m}\hat{L}_z$$

The eigenvalue of magnetic moment

$$\mu_l = -\frac{e\hbar}{2m}\sqrt{l(l+1)}$$

$$\mu_z = -\frac{e\hbar}{2m}m_l$$

Bohr magneton

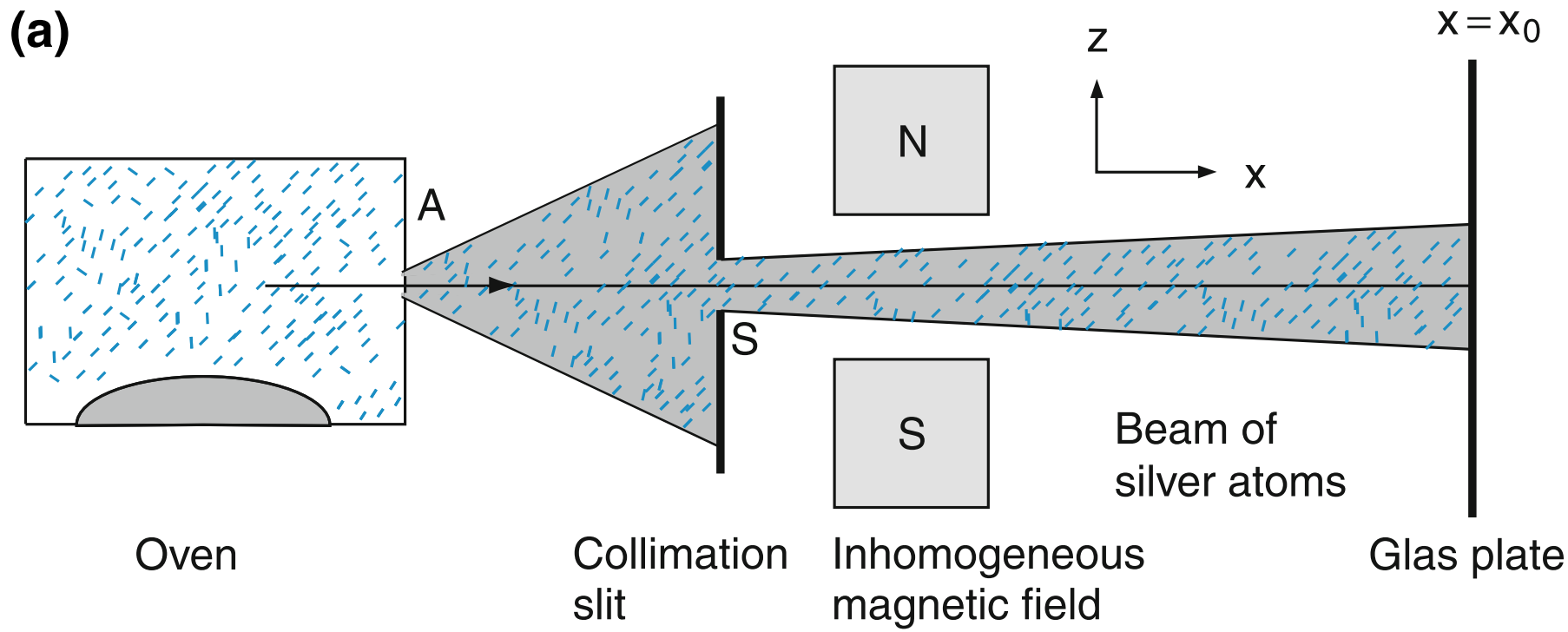
$$\mu_B = \frac{e\hbar}{2m}$$

The energy of the orbiting electron in a magnetic field

$$E_B = -\mu_z B = \mu_B m_l B$$

4.2 Stern-Gerlach experiment

In 1922 O. Stern and W. Gerlach reported the results of an experiment that clearly showed evidence for space quantization.



Stern-Gerlach experiment

They started in 1919 and after many unsuccessful efforts and following improvements to their beam apparatus, the two researchers were finally rewarded with an unexpected result in 1921

They chose as test objects silver atoms because these atoms could be detected on a glass plate, where they condensate and form a thin layer with an optical transparency depending on the thickness of the layer and therefore the number of incident atoms.

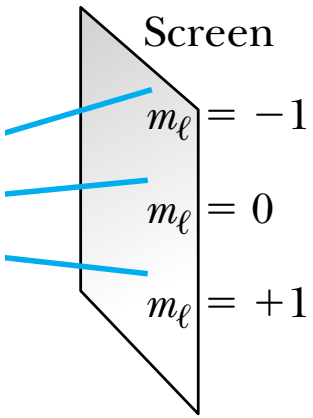
Stern-Gerlach experiment

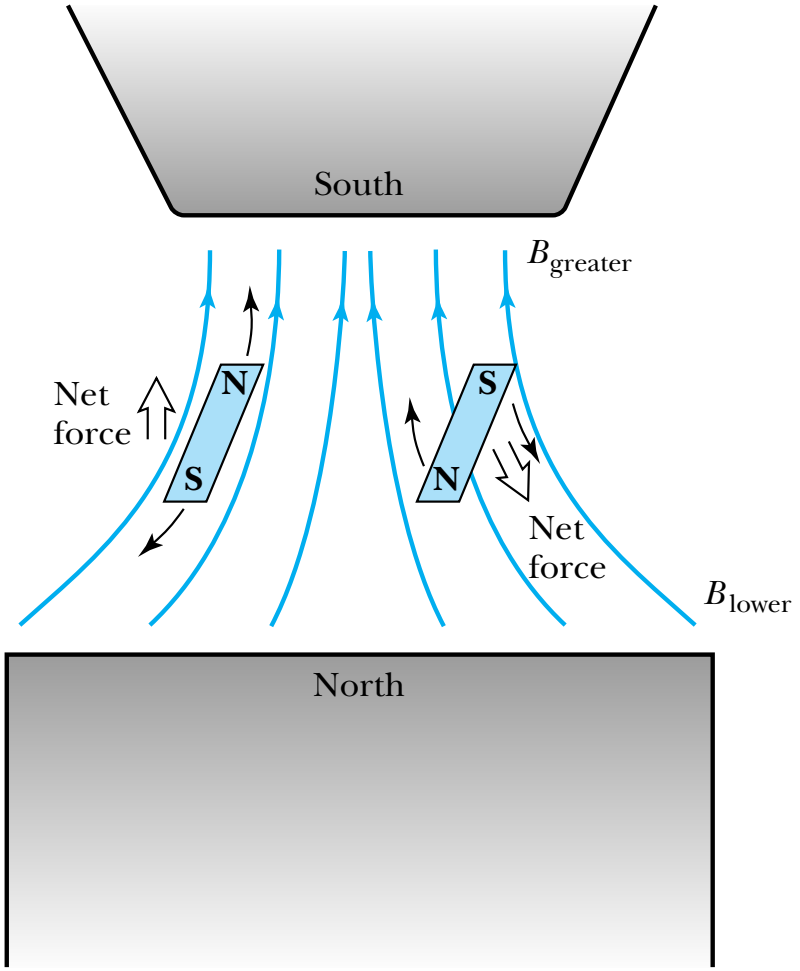
If an external magnetic field is inhomogeneous, the force on the particles is

$$F_z = -\frac{dE_B}{dz} = \mu_z \frac{dB}{dz}$$

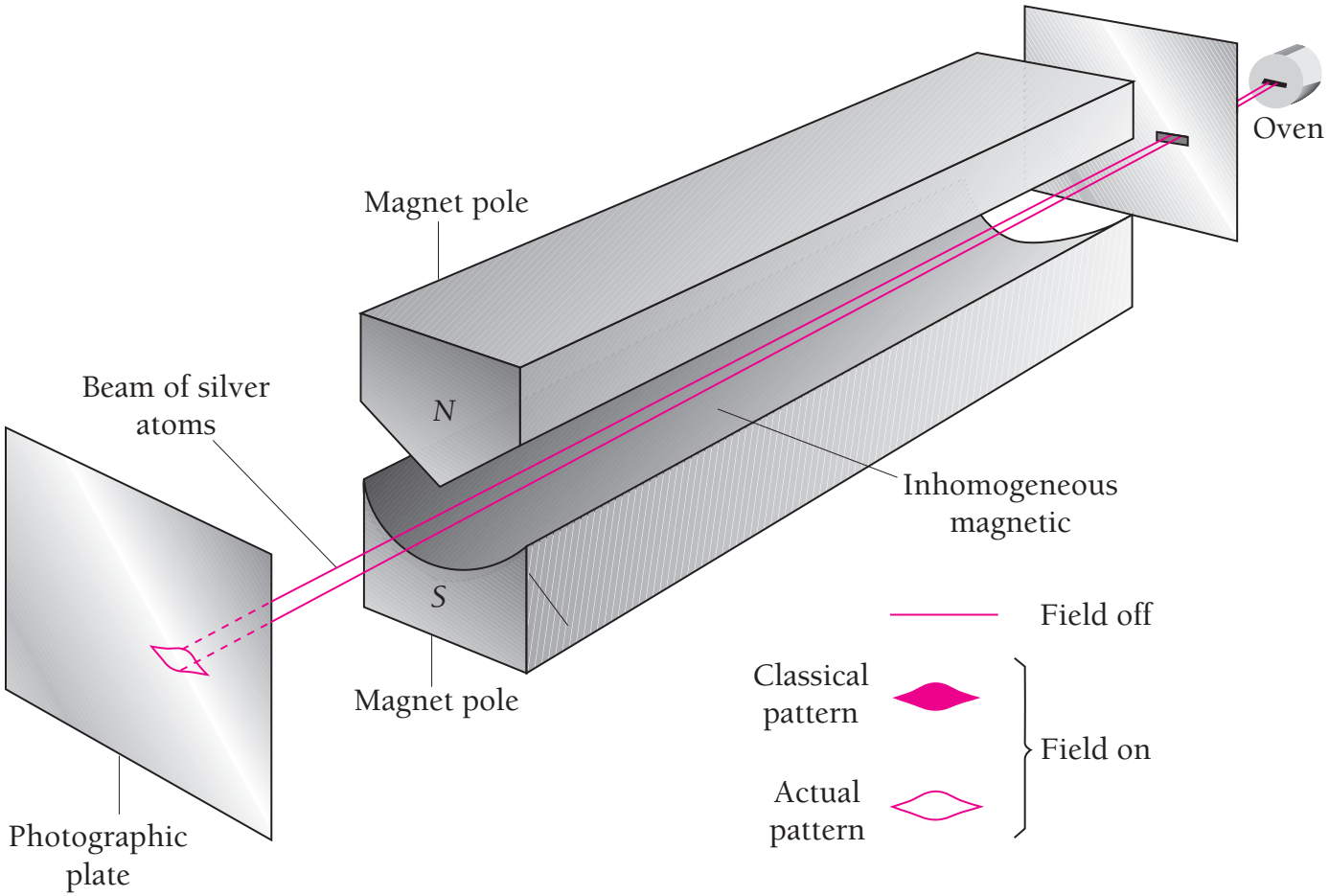
When l=1

$$\mu_z = -1,\ 0,\ 1\mu_B$$





Stern-Gerlach experiment



The acceleration of the atom along the z direction is

$$a_z = F_z/m.$$

The separation of the atom along the z direction due to this acceleration is

$$d = a_z t^2/2.$$

The time that the atom spends within the inhomogeneous field is

$$t = \Delta x/v_x$$

where Δx is the length of the inhomogeneous field, and v_x is the constant speed of the atom within the field. The separation d is therefore found from

$$d = \frac{1}{2}a_z t^2 = \frac{1}{2}\left(\frac{F_z}{m}\right)t^2 = \frac{1}{2m}\left(\mu_z \frac{dB}{dz}\right)\left(\frac{\Delta x}{v_x}\right)^2$$

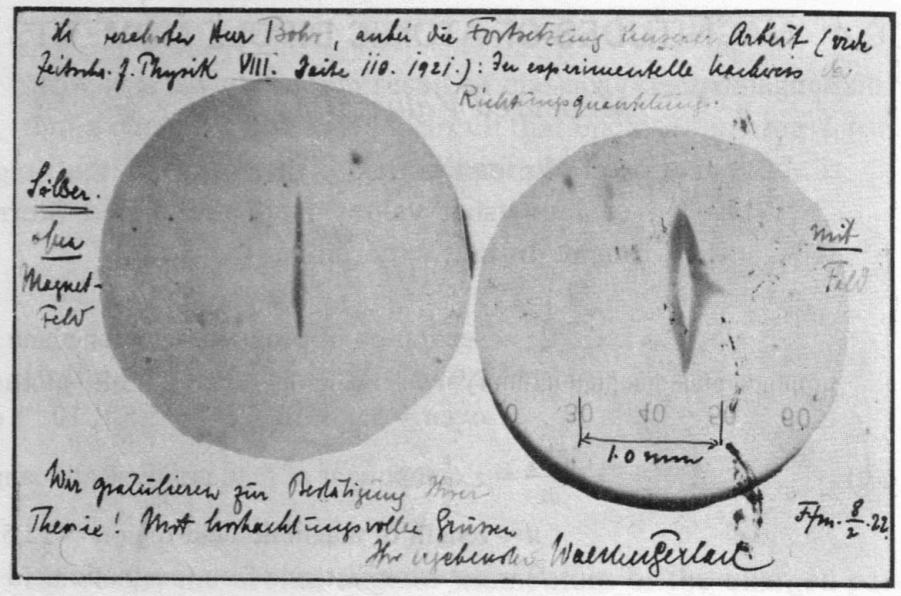
$$\tfrac{1}{2}m\langle v_x^2\rangle = \tfrac{3}{2}kT.$$

Stern-Gerlach experiment



Stern-Gerlach experiment

Stern and Gerlach performed their experiment with silver atoms and observed two distinct lines, not three.

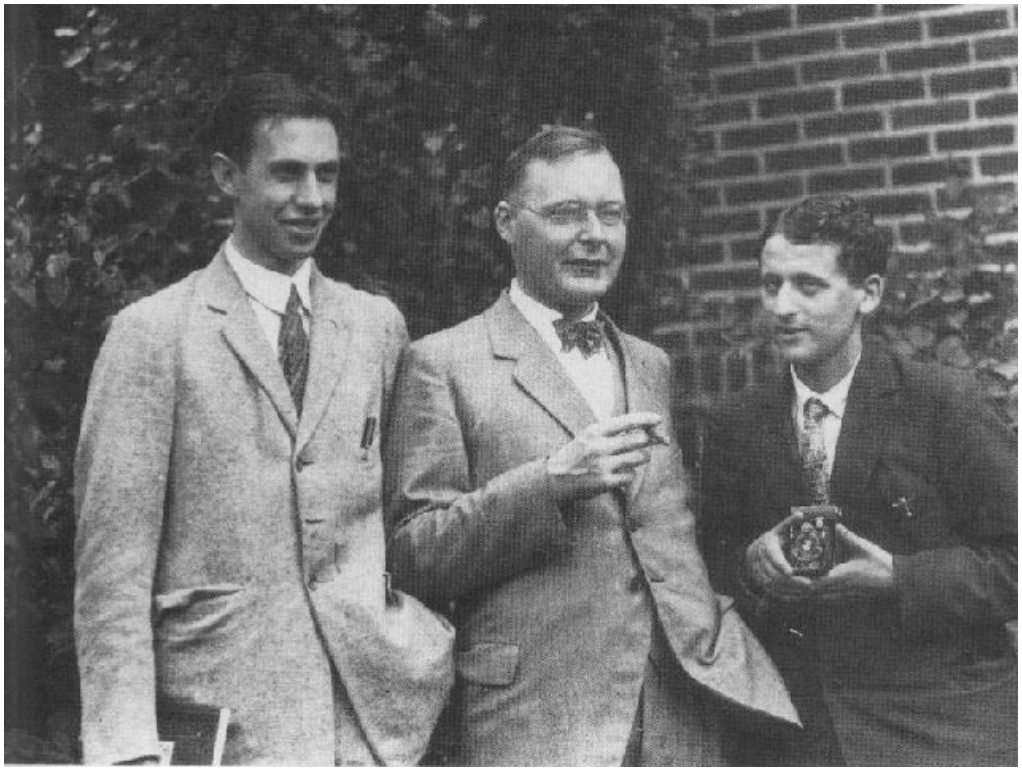


This was clear evidence of space quantization, although the number of m_l states is always odd and should have produced an odd number of lines if the space quantization were due to the magnetic quantum number m_l.

4.3 Intrinsic Spin

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In 1925 Samuel Goudsmit and George Uhlenbeck, two young physics graduate students in Holland, proposed that the electron must have an intrinsic angular momentum and therefore a magnetic moment



Intrinsic Spin

In order to achieve the angular momentum needed, Paul Ehrenfest showed that the surface of the spinning electron (or electron cloud) would have to be moving at a velocity greater than the speed of light!

If such an intrinsic angular momentum exists, we must regard it as a purely quantum-mechanical result.

To explain experimental data, Goudsmit and Uhlenbeck proposed that the electron must have an intrinsic spin quantum number

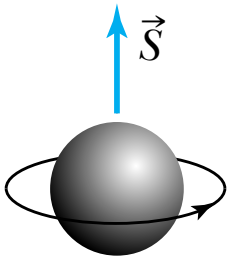
$$s = 1/2$$

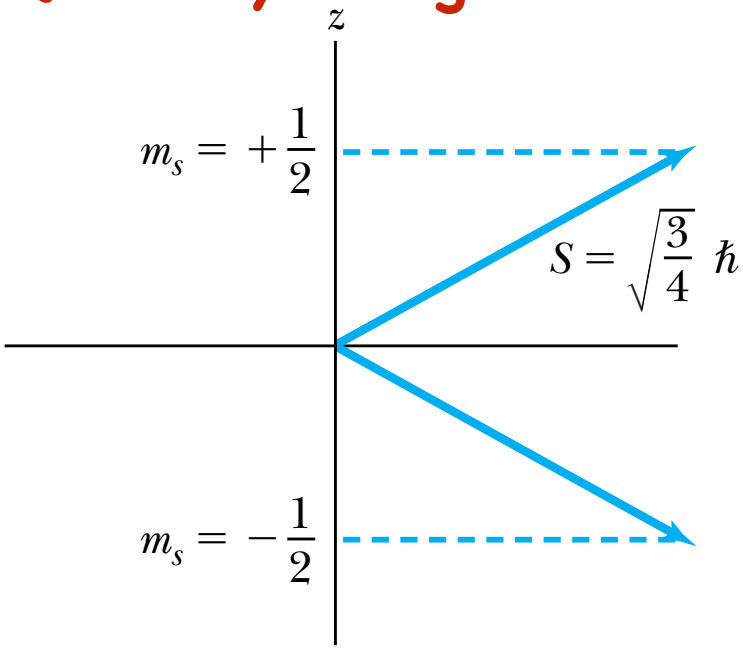
Intrinsic Spin

The magnetic spin quantum number m_s has only two values

$$m_s = \pm 1/2$$

The electron's spin will be oriented either "up" or "down" in a magnetic field, and the electron can never be spinning with its magnetic moment m_s exactly along the z axis





Intrinsic Spin

The intrinsic spin angular momentum

$$|\vec{S}| = \sqrt{s(s+1)}\hbar = \sqrt{3/4}\hbar$$

The magnetic moment of intrinsic spin

$$\vec{\mu}_s = -g_s \mu_B \vec{S}/\hbar = -2\mu_B \vec{S}/\hbar$$

where g_s is the Lande factor and

$$g_s = 2$$

The z component of spin magnetic moment

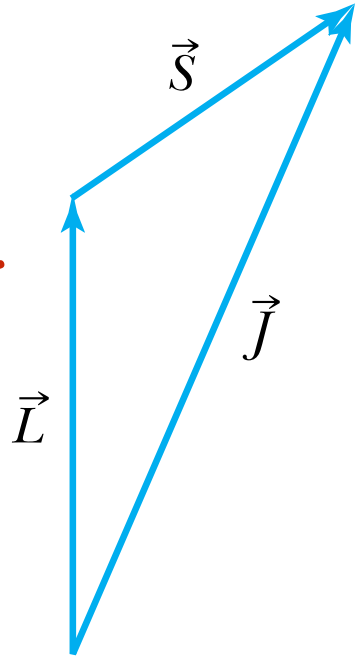
$$\mu_{s,z} = -2m_s \mu_B$$

where

$$m_s = \pm 1/2$$

If an atom has an orbital angular momentum and a spin angular momentum, these angular momenta combine to produce a total angular momentum,

$$\vec{J} = \vec{L} + \vec{S}$$



Because L, L_z, S, and S_z are quantized, the total angular momentum and its z component J_z are also quantized

$$J = \sqrt{j(j+1)}\hbar$$

$$J_z = m_j \hbar$$

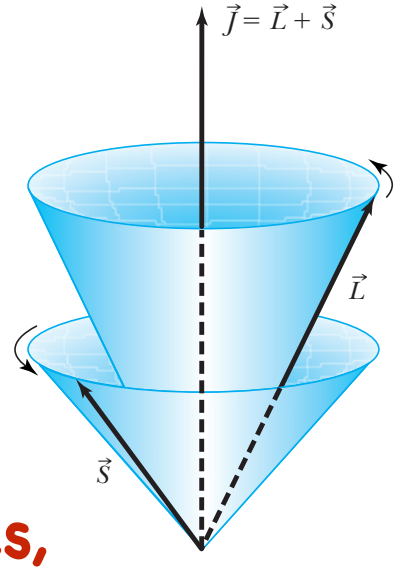
Because m_l is integral and m_s is half-integral, m_j will always be half-integral.

The quantization of the magnitudes of various angular are all similar,

$$L = \sqrt{l(l+1)}\hbar$$

$$S = \sqrt{s(s+1)}\hbar$$

$$J = \sqrt{j(j+1)}\hbar$$



The total angular momentum quantum number for the single electron can only have the values,

$$j = l \pm s$$

$$= l \pm \frac{1}{2}$$

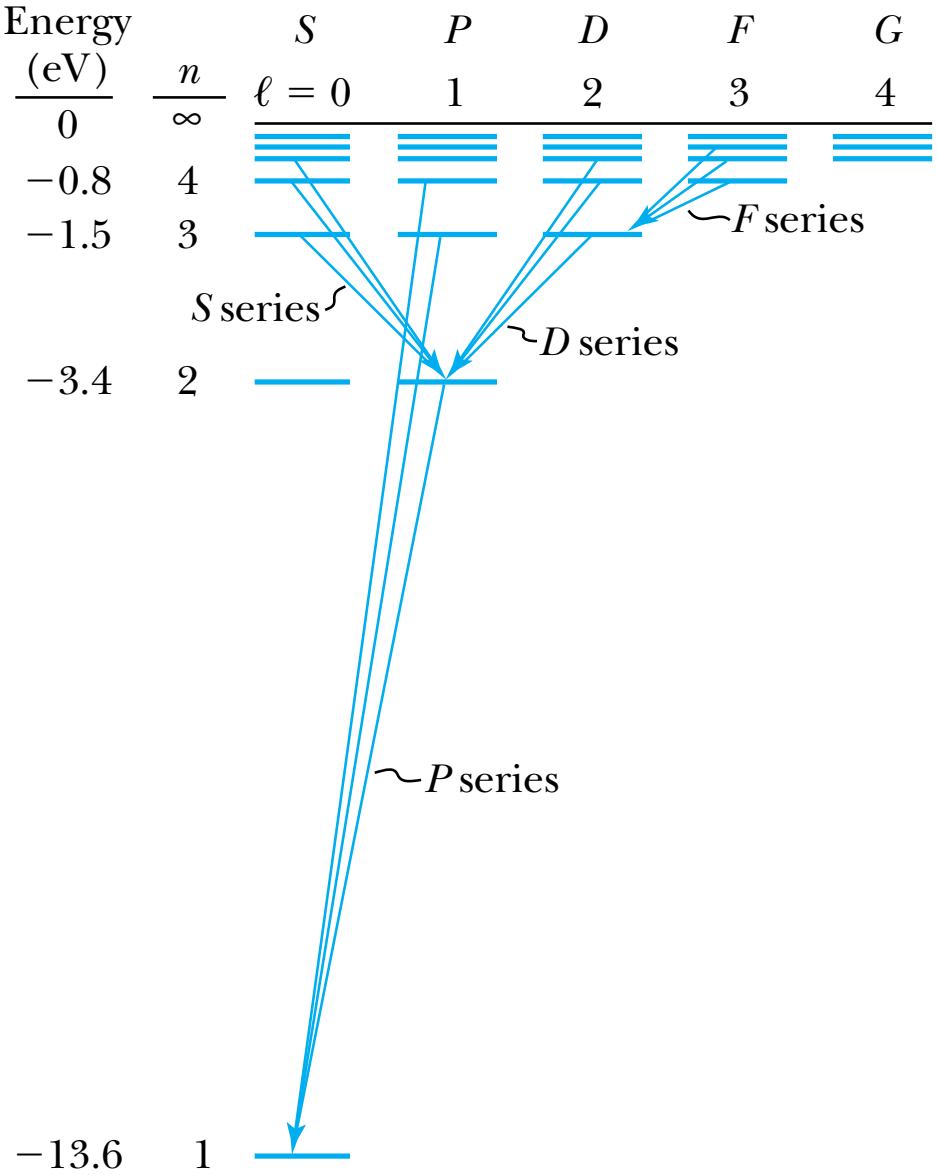
The notation commonly used to describe these states

$$^{2S+1}L_J$$

$$L = 0, 1, 2, 3, \ldots$$

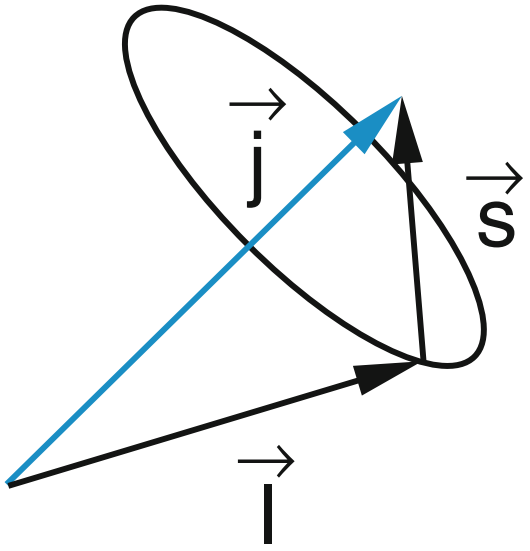
$$= S, P, D, F, \ldots$$

Total angular momentum of single electron



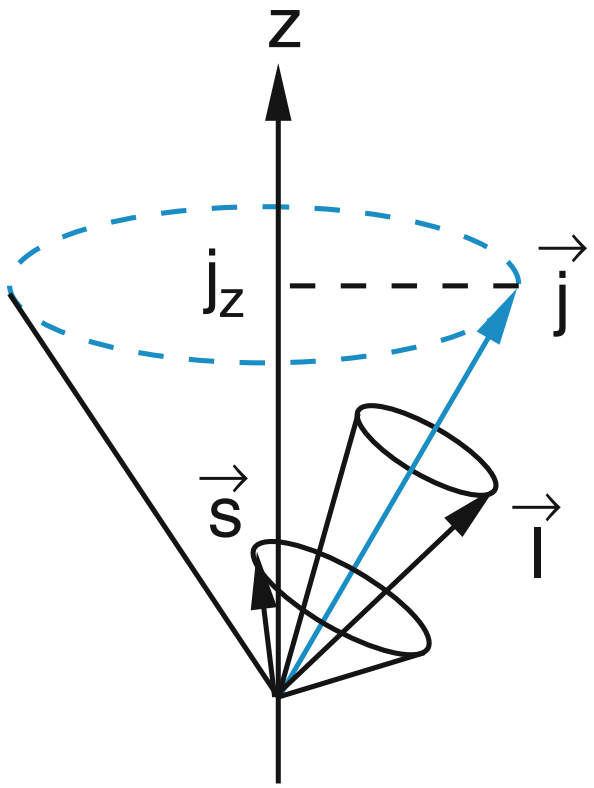
Total magnetic moment

Without a magnetic field, the total angular momentum j = l + s of the electron in a Coulomb field (central force field) is constant, which means that its orientation in space and its absolute value are constant, independent of time.



Total magnetic moment

In an external homogeneous magnetic field B = {0, 0, B_z}, the magnetic moment μ_j, and therefore j, precess around the field axis with constant components μ_z and j_z



Total magnetic moment

If the coupling between orbital angular momentum and spin is stronger than the coupling of μ_l and μ_s to the external field. The spin-orbit coupling is still valid and the absolute value |j| of the total angular momentum

$$\boldsymbol{j} = \boldsymbol{l} + \boldsymbol{s} \quad \text{with} \quad |\boldsymbol{j}| = \sqrt{j(j+1)}h$$

is conserved in the external magnetic field. Its direction is, however, no longer space-fixed because the magnetic moment $\mu_j = \mu_l + \mu_s$, which is related to j, experiences a torque

$$\boldsymbol{D} = \boldsymbol{\mu}_j \times \boldsymbol{B}.$$

Total magnetic moment

The magnetic moments of orbital motion and spin of the electron are:

$$\boldsymbol{\mu}_l = -(\mu_B/\hbar)\boldsymbol{l} \quad \text{and} \quad \boldsymbol{\mu}_s = -g_s(\mu_B/\hbar)\boldsymbol{s}.$$

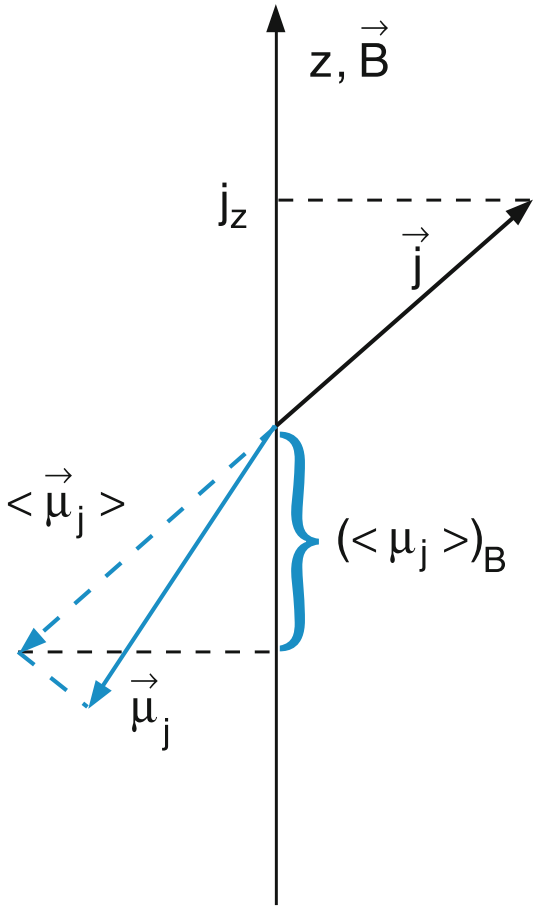
Because of the Lande factor $g_s \approx 2$ for the spin moment μ_s, the total magnetic moment

$$\boldsymbol{\mu}_j = \boldsymbol{\mu}_l + \boldsymbol{\mu}_s = -(\mu_B/\hbar)(\boldsymbol{l} + g_s\boldsymbol{s})$$

is for l > 0 no longer parallel to the total angular momentum j = l + s!

Total magnetic moment

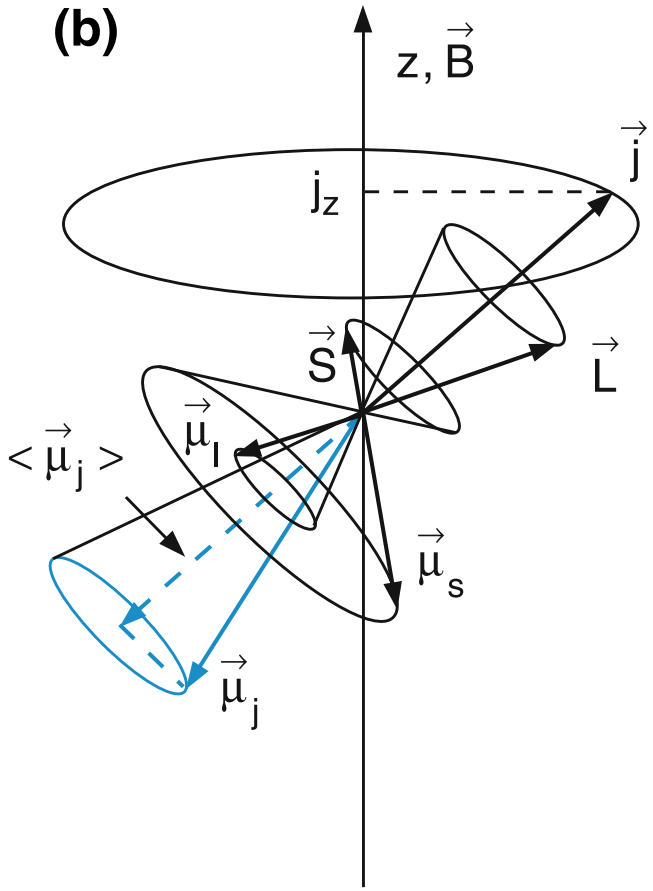
Without an external field the absolute value and the direction of j is constant in time. Since the vector s precesses around the axis of the internal magnetic field produced by the orbital movement of the electron and μ_j is not parallel to j, μ_j has to precess around the direction of the space-fixed vector j. The time average $\langle \mu_j \rangle$ of μ_j is then the projection of μ_j onto j.



Total magnetic moment

This gives

$$\langle \boldsymbol{\mu}_j \rangle = \frac{\boldsymbol{\mu}_j \cdot \boldsymbol{j}}{|\boldsymbol{j}|^2}\boldsymbol{j} = -\frac{e}{2m_e}\left(\frac{\boldsymbol{l}\cdot\boldsymbol{j}}{|\boldsymbol{j}|^2} + g_s\frac{\boldsymbol{s}\cdot\boldsymbol{j}}{|\boldsymbol{j}|^2}\right)\boldsymbol{j}.$$



Total magnetic moment

From j = l + s follows

$$\boldsymbol{l}\cdot\boldsymbol{j} = \frac{1}{2}[\boldsymbol{j}^2 + \boldsymbol{l}^2 - \boldsymbol{s}^2]$$

$$= \frac{1}{2}[j(j+1) + l(l+1) - s(s+1)]\hbar^2$$

and similar from l = j – s:

$$\boldsymbol{s}\cdot\boldsymbol{j} = \frac{1}{2}[j(j+1) + s(s+1) - l(l+1)]\hbar^2.$$

We can therefore write for the component of $\langle\mu_j\rangle$ in the direction of j with $g_s \approx 2$

$$\langle\boldsymbol{\mu}_j\rangle_j = -\frac{3j(j+1) + s(s+1) - l(l+1)}{2j(j+1)}\mu_B\hat{j} = -g_j\,\mu_B\hat{j}$$

Total magnetic moment

The Lande factor g_j is defined here as

$$g_j = 1 + \frac{j(j+1) + s(s+1) - l(l+1)}{2j(j+1)}.$$

For s = 0 (pure orbital magnetism) it follows j = l and we obtain g_j =1. For l=0 (pure spin magnetism) is j=s and therefore $g_j \approx 2$.

In the z direction,

$$\langle \mu_j \rangle_z = -m_j g_j \mu_B$$

$$\Delta E_{m_j} = -\langle \mu_j \rangle_z B = m_j g_j \mu_B B.$$

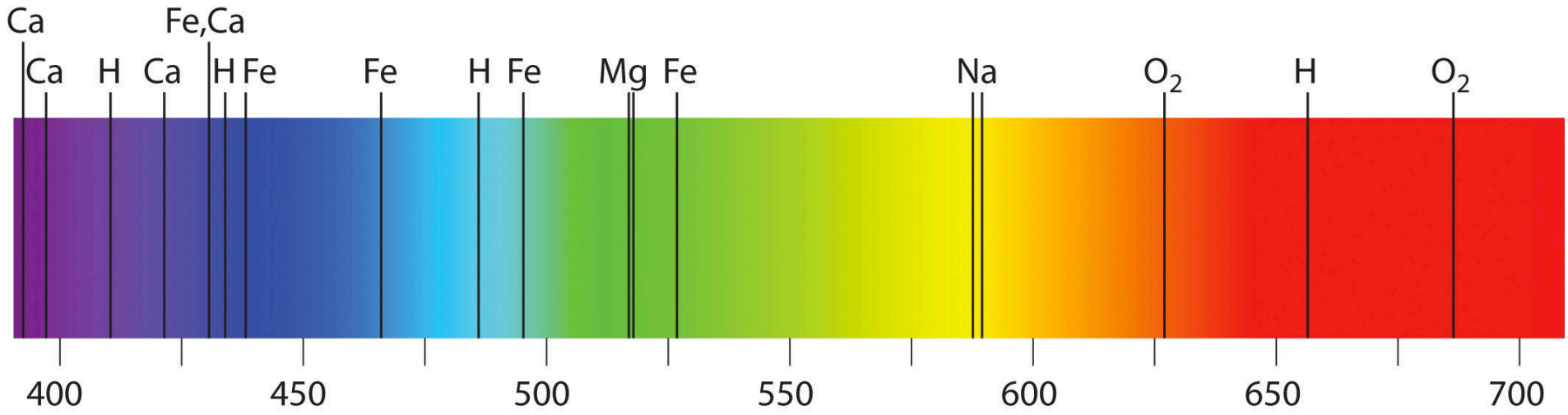
Total magnetic moment

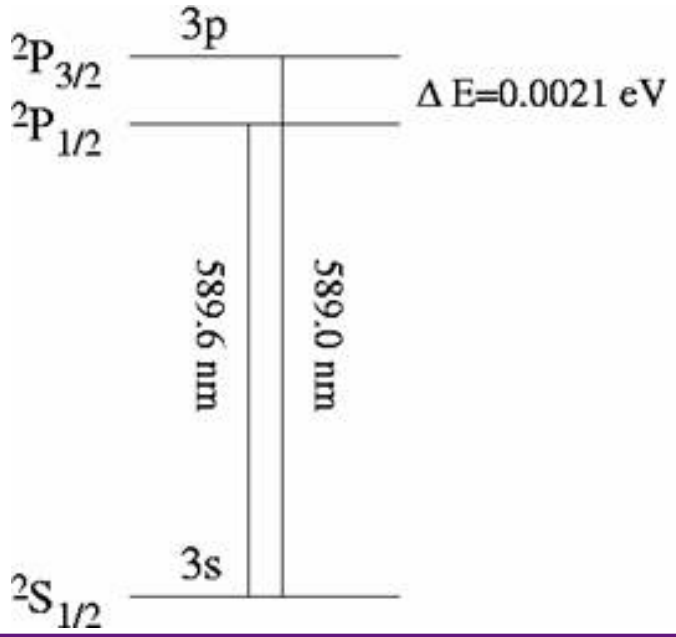
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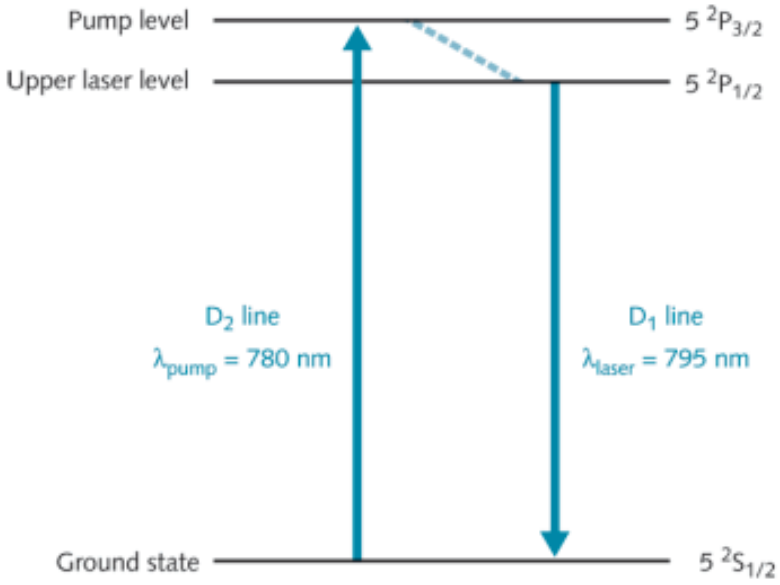
Lande factor of different state

States	g_j	$m_j g_j$
$^2S_{1/2}$	2	±1
$^2P_{1/2}$	2/3	±1/3
$^2P_{3/2}$	4/3	±2/3, ±6/3
$^2D_{3/2}$	4/5	±2/5, ±6/5
$^2D_{5/2}$	6/5	±3/5, ±9/5, ±15/5

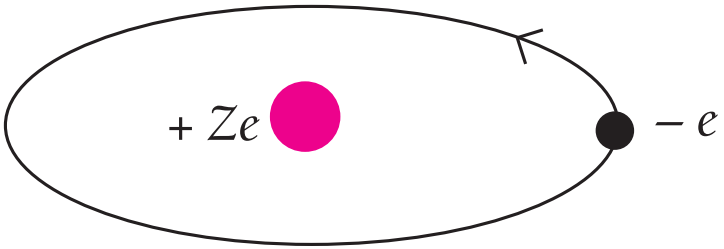
4.5 Spin-orbit coupling

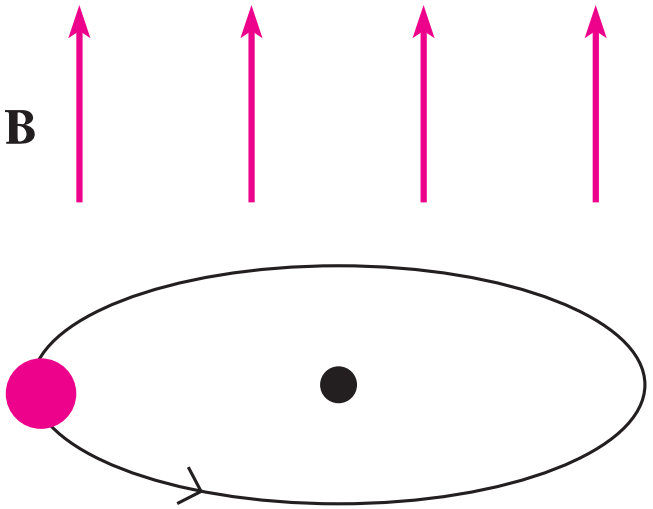






This classical picture indicates that the orbiting proton creates a magnetic field at the position of the electron.





Spin-orbit coupling

An electron moving through an electric field E experiences an effective magnetic field B given by

$$\vec{B} = -\frac{1}{c^2}\vec{v} \times \vec{E}$$

Since, the electric field is

$$\vec{E} = \frac{1}{e}\frac{\partial V}{\partial r}\frac{\vec{r}}{r}$$

We have

$$\vec{B} = \frac{1}{m_e c^2}\frac{1}{er}\frac{\partial V}{\partial r}\vec{L}$$

The potential of the electron's magnetic moment with the orbital field gives

$$U = -\vec{\mu}_s \cdot \vec{B}$$

$$= g_s \mu_B \frac{1}{m_e c^2 er}\frac{\partial V}{\partial r}\vec{L} \cdot \vec{S}$$

Spin-orbit coupling

With Coulomb potential and relativistic correction that arises because we are calculating the magnetic field in a frame of reference that is not stationary but rotates as the electron moves about the nucleus,

$$U = \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{2m_e^2c^2r^3}\vec{L}\cdot\vec{S}$$

The expectation value of this potential gives an energy change of

$$E_{so} = \frac{1}{8\pi\varepsilon_0 m_e^2 c^2}\langle\frac{1}{r^3}\rangle\langle\vec{L}\cdot\vec{S}\rangle$$

The expectation value of spin-orbit coupling

$$\langle \vec{L} \cdot \vec{S} \rangle = \frac{j(j+1) - l(l+1) - s(s+1)}{2} \hbar^2$$

The expectation value of radii

$$\langle \frac{1}{r^3} \rangle = \frac{Z^3}{n^3 l(l+1/2)(l+1) a_1^3}$$

Thus the spin-orbit interaction produces a shift in energy of

$$E_{so} = \frac{Z^4 e^2 \hbar^2}{16\pi\varepsilon_0 m_e^2 c^2 a_1^3} \frac{j(j+1) - l(l+1) - s(s+1)}{n^3 l(l+1/2)(l+1)}$$

Spin-orbit coupling

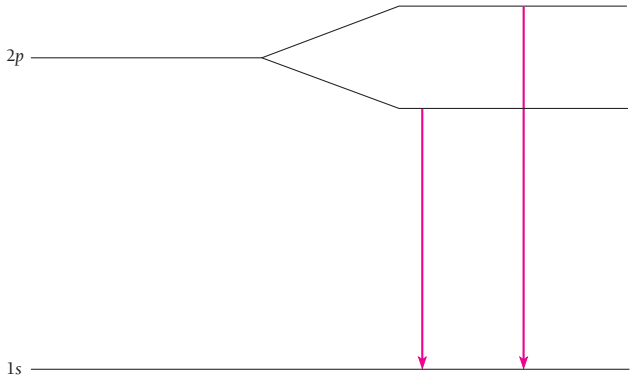
A single electron has s =1/2 so, for each l, its total angular momentum quantum number j has two possible values:

$$j = l \pm \frac{1}{2}$$

We find that the energy interval between these levels:

$$\Delta E_{so} = \frac{Z^4 e^2 \hbar^2}{8\pi\varepsilon_0 m_e^2 c^2 a_1^3} \frac{1}{n^3 l(l+1)}$$

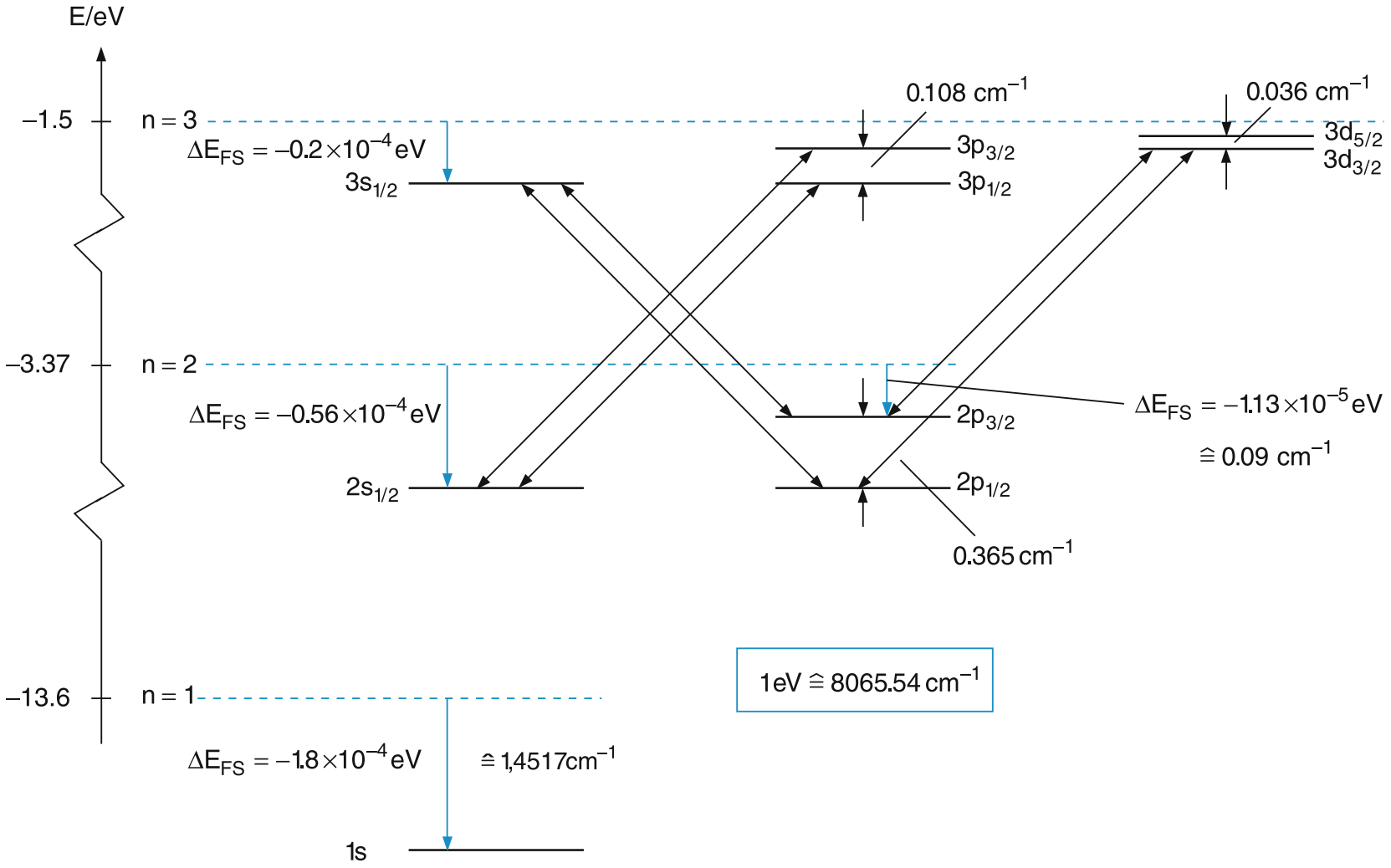
$$= \frac{(\alpha Z)^4}{2n^3 l(l+1)} E_0$$



The energy interval of 2p state of hydrogen

$$\Delta E_{so} = 4.53 \times 10^{-5} \text{ eV}$$

Hydrogen fine structure spectrum



$$E_{n,j} = E_n\left[1 + \frac{Z^2\alpha^2}{n}\left(\frac{1}{j+1/2} - \frac{3}{4n}\right)\right],$$

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The nuclear spin

$$|\boldsymbol{I}| = \sqrt{I(I+1)}\hbar$$

The projection of I onto the z-axis is

$$I_z = m_I \hbar \quad \text{with} \quad -I \leq m_I \leq +I.$$

A magnetic moment is connected with the nuclear spin

$$\mu_{\mathrm{N}} = \gamma_{\mathrm{K}} I.$$

The unit of the nuclear magnetic moment is the nuclear magneton

$$\mu_{\mathrm{K}} = \frac{e}{2m_{\mathrm{p}}}\hbar = \frac{m_{\mathrm{e}}}{m_{\mathrm{p}}}\mu_{\mathrm{B}} = \frac{\mu_{\mathrm{B}}}{1836}$$

$$= 5.05 \times 10^{-27}\,\mathrm{JT}^{-1}$$

Hyperfine structure

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The magnetic moment of any atomic nucleus can be written in units of the nuclear magneton as

$$\mu_{\mathrm{N}} = \gamma_{\mathrm{K}} I = g_{\mathrm{N}} \frac{\mu_{\mathrm{K}}}{\hbar} I,$$

The where the dimensionless factor g_N is called the nuclear g-factor.

The magnetic moment of the proton is $\mu_I(p) = 2.79\mu_K$ and is determined by the movements and charges of the three quarks (u,u,d) inside the proton.

Hyperfine structure

The nuclear magnetic moment gives two contributions to the shift and splitting of energy levels of the atomic electrons:

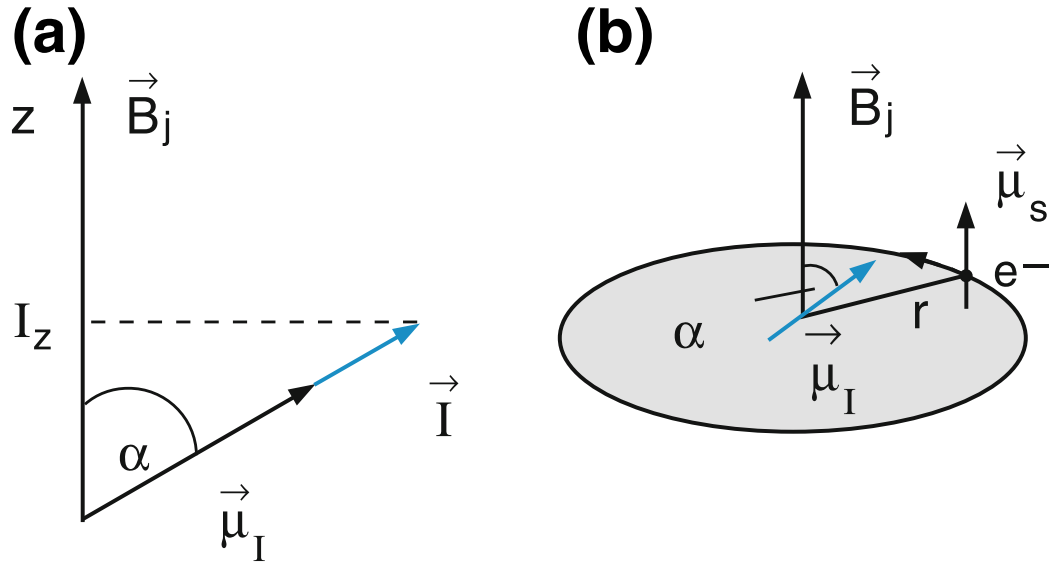
a) The interaction of the nuclear magnetic moment μ_N with the magnetic field produced by the electrons at the nucleus (Zeeman effect of μ_N in the internal magnetic field produced by the electrons).

b) The interaction of the electronic magnetic moment μ_j with the nuclear moment μ_N (magnetic dipole-dipole interaction).

Hyperfine structure

The potential energy of the nuclear magnetic moment μ_N in the magnetic field produced by the electron at the location of the nucleus is

$$E_{\text{pot}}(I, j) = -\boldsymbol{\mu}_{\text{N}} \cdot B_{\text{int}} = -|\boldsymbol{\mu}_{\text{N}}| B_j \cos(\angle \boldsymbol{j}, \boldsymbol{I}).$$



$$E_{HFS} = -\vec{\mu}_I \cdot \vec{B}_j$$

$$\vec{B} = \vec{B}(\vec{\mu}_s, \vec{\mu}_I, r)$$

Hyperfine structure

Introducing the total angular momentum F = j+I of the atom as the vector sum of the total electronic angular momentum j = l + s and the nuclear spin I, we obtain

$$\cos\angle(\boldsymbol{j},\boldsymbol{I}) = \frac{\boldsymbol{j}\cdot\boldsymbol{I}}{|\boldsymbol{j}||\boldsymbol{I}|} = \frac{1}{2}\frac{F(F+1)-j(j+1)-I(I+}{\sqrt{j(j+1)I(I+1)}} .$$

The hyperfine energy of the H atom is then

$$\Delta E_{\mathrm{HFS}} = \frac{A}{2}[F(F+1)-j(j+1)-I(I+1)],$$

where the hyperfine constant

$$A = \frac{g_{\mathrm{N}}\mu_{\mathrm{K}}B_j}{\sqrt{j(j+1)}}$$

Hyperfine structure

Each energy level $E_{n,j,l}$, splits into hyperfine components, due to the interaction between nuclear magnetic moment and electronic magnetic moments. The energy of these components is

$$E_{\mathrm{HFS}} = E_{n,l,j} + \frac{1}{2}A[F(F+1) - j(j+1) - I(I+1)].$$

For the H atom, with a proton as the nucleus, the experiments give the values

$$I = 1/2, \quad g_I = +5.58 \Rightarrow (\mu_{\mathrm{N}})_z = \pm 2.79\mu_{\mathrm{K}}$$

Hyperfine structure

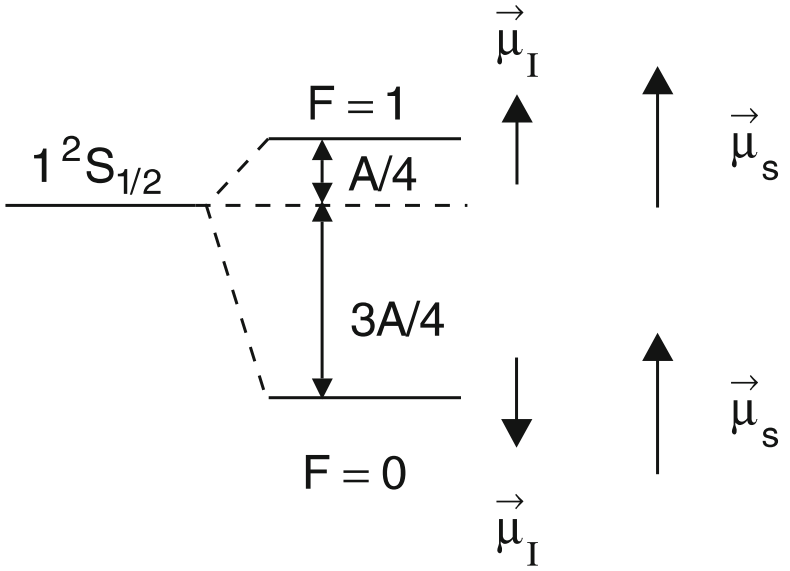
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For the ground state $^2S_{1/2}$ is j = 1/2, I = 1/2 ⇒ F = 0 or F = 1.

This gives the two hyperfine components

$$E_{\mathrm{HFS}}(F=0) = E_{1,0,1/2} - \frac{3}{4}A$$

$$E_{\mathrm{HFS}}(F=1) = E_{1,0,1/2} + \frac{1}{4}A,$$



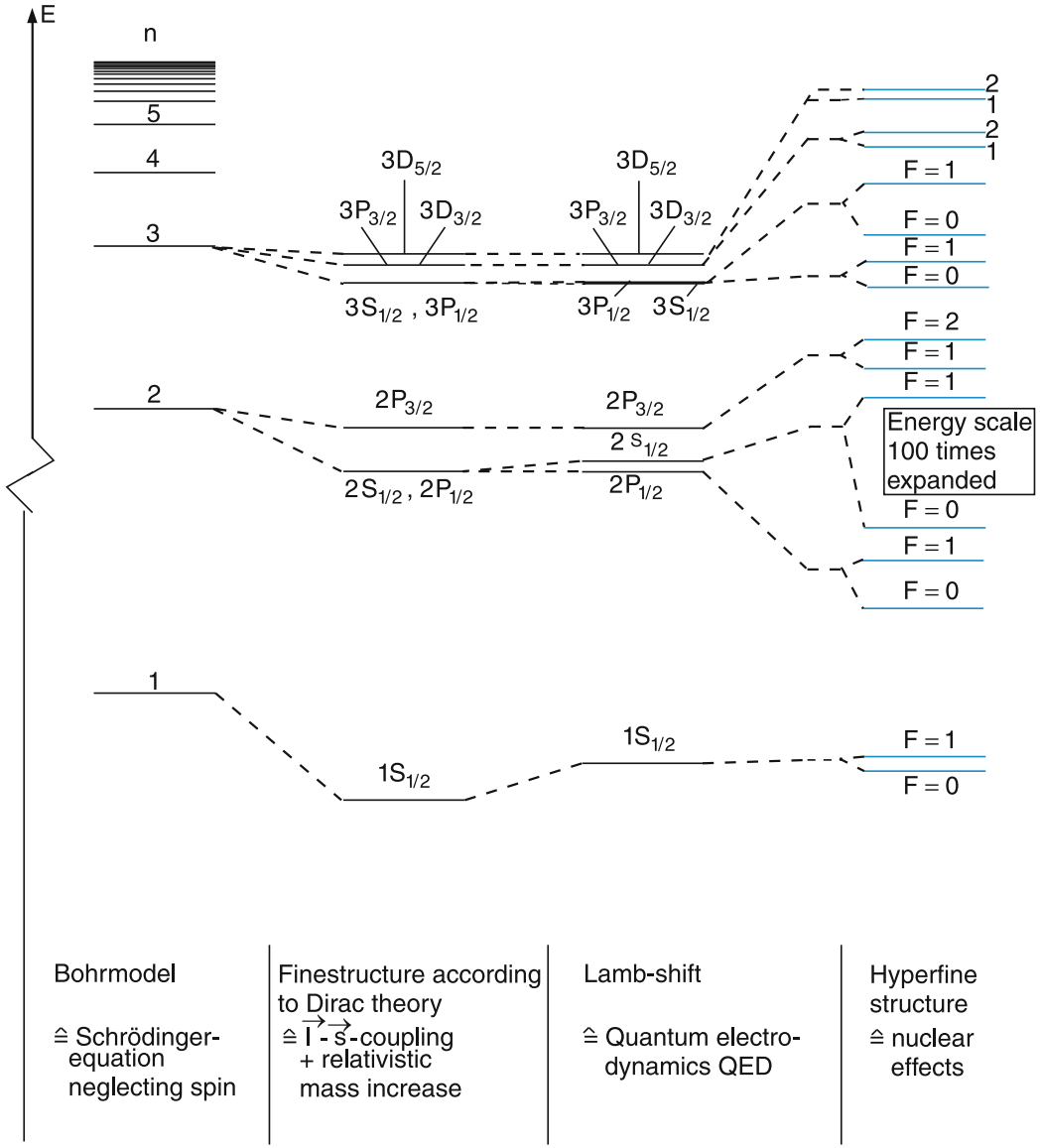
with the separation

$$\Delta E = A = 5.8 \cdot 10^{-6}\,\mathrm{ev}$$

and

$$\Delta\lambda = 21\,\mathrm{cm}$$

Complete Description of H



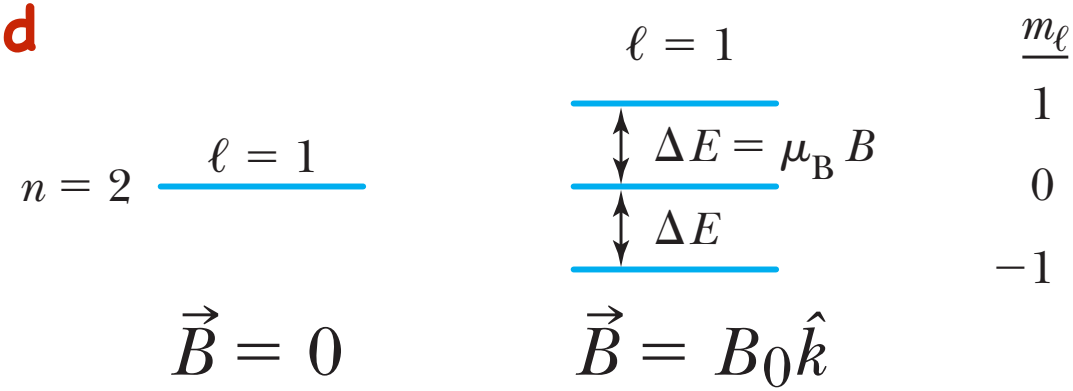
Zeeman effect

In a magnetic field, then, the energy of a particular atomic state depends on the value of m_l as well as on that of n

$$E_B = -\mu_z B = \mu_B m_l B$$

Each (degenerate) atomic level of given l is split into 2l+1 different energy states according to the value of m_l

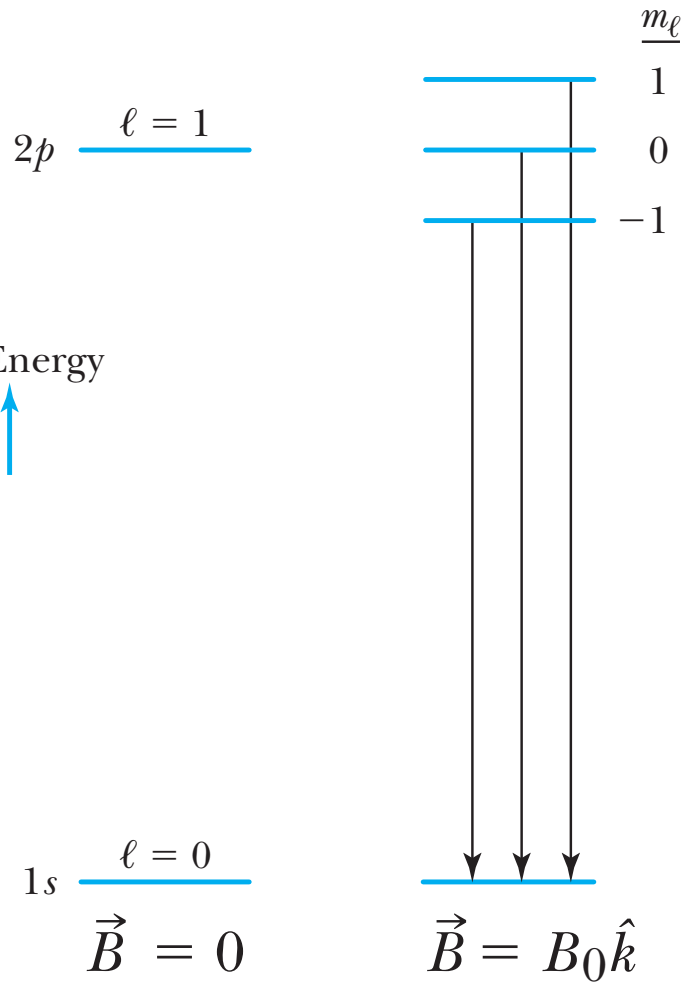
The energy degeneracy of a given nl level is removed by a magnetic field



If the degenerate energy of a state is given by E_0, the three different energies in a magnetic field B for a l=1 state are

$$E = E_0,\ E_0 \pm \mu_B B$$

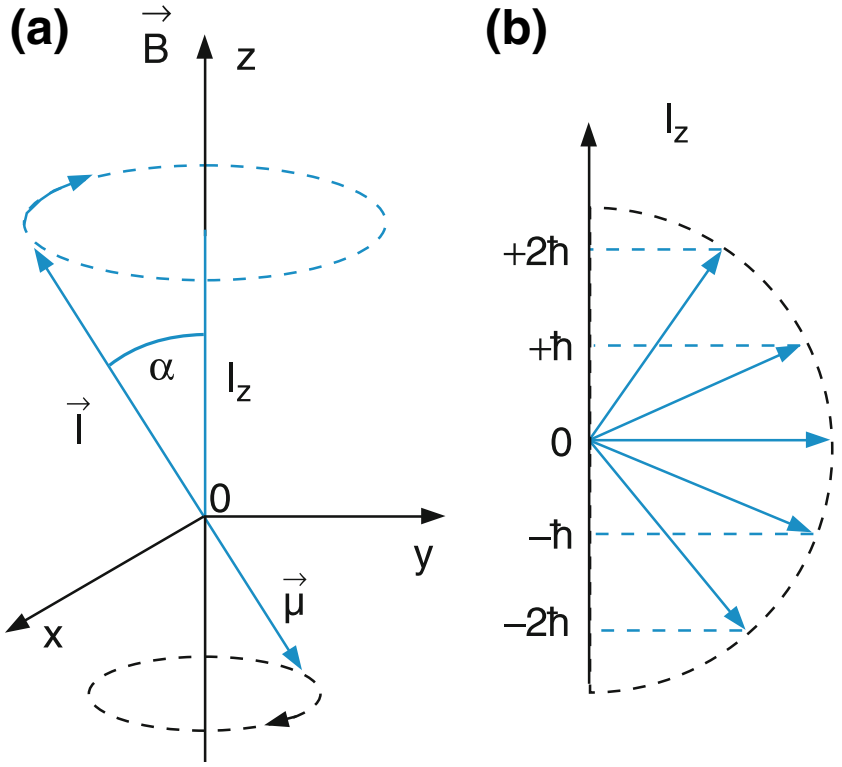
m_ℓ	Energy
1	$E_0 + \mu_B B$
0	E_0
−1	$E_0 - \mu_B B$



Zeeman effect

(a) Classical model of orbital angular momentum precessing around the field axis.

(b) Possible orientations of l and projections m in the quantum mechanical description



Zeeman effect

Because m_l can have the 2l+1 values, a state of given orbital quantum number l is split into 2l+1 substates that differ in energy by $\mu_B B$ when the atom is in a magnetic field.

However we expect a spectral line from a transition between two states of different l to be split into only three components for selection rules

$$\Delta m_l = 0, \pm 1$$

due to the spin of phonon as one.

Zeeman effect

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The normal Zeeman effect consists of the splitting of a spectral line of frequency into three components whose frequencies are

$$\nu_1 = \nu_0 - \frac{e}{4\pi m} B$$

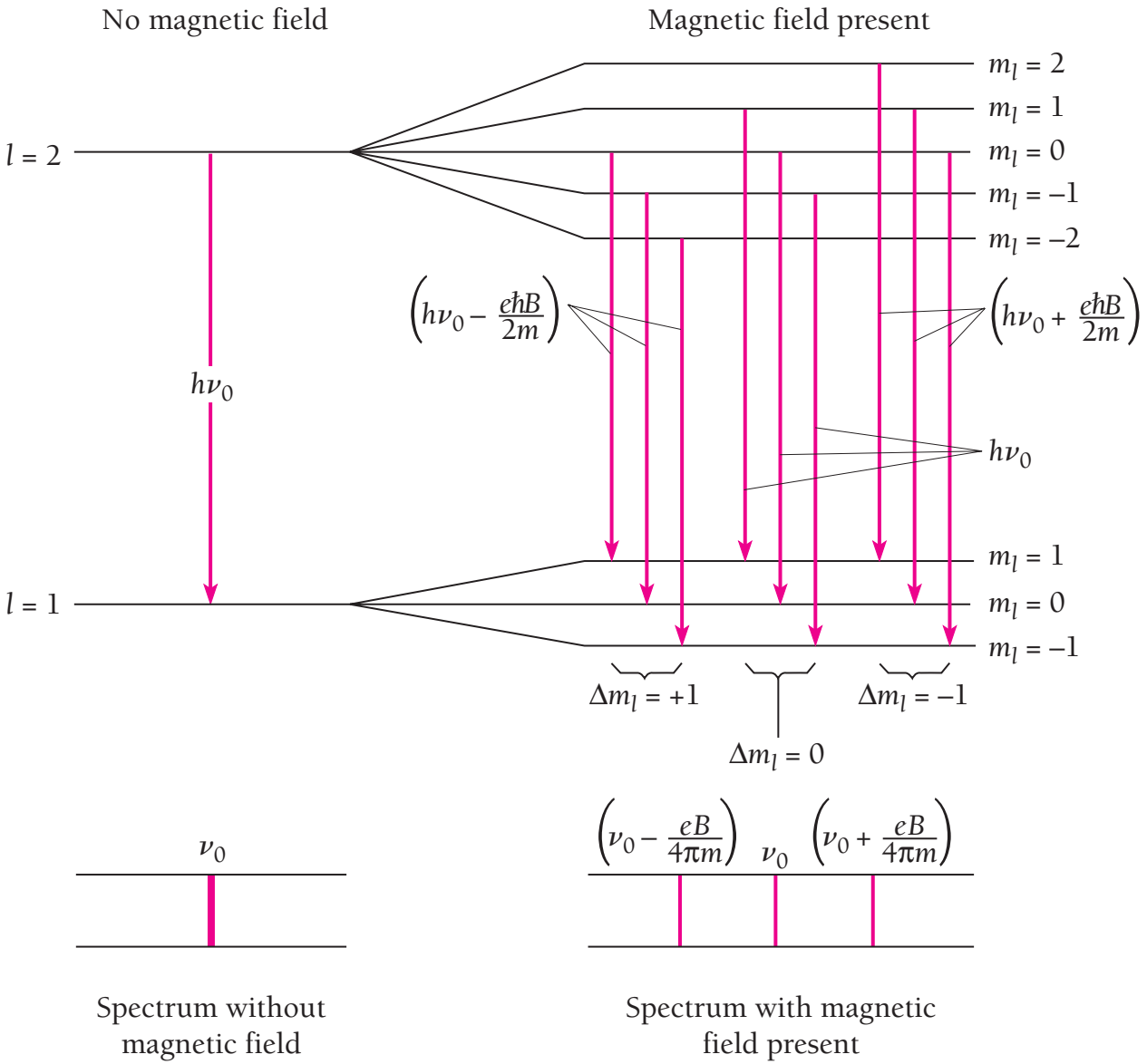
$$\nu_2 = \nu_0$$

$$\nu_3 = \nu_0 + \frac{e}{4\pi m} B$$

where, the Larmor frequency is defined as

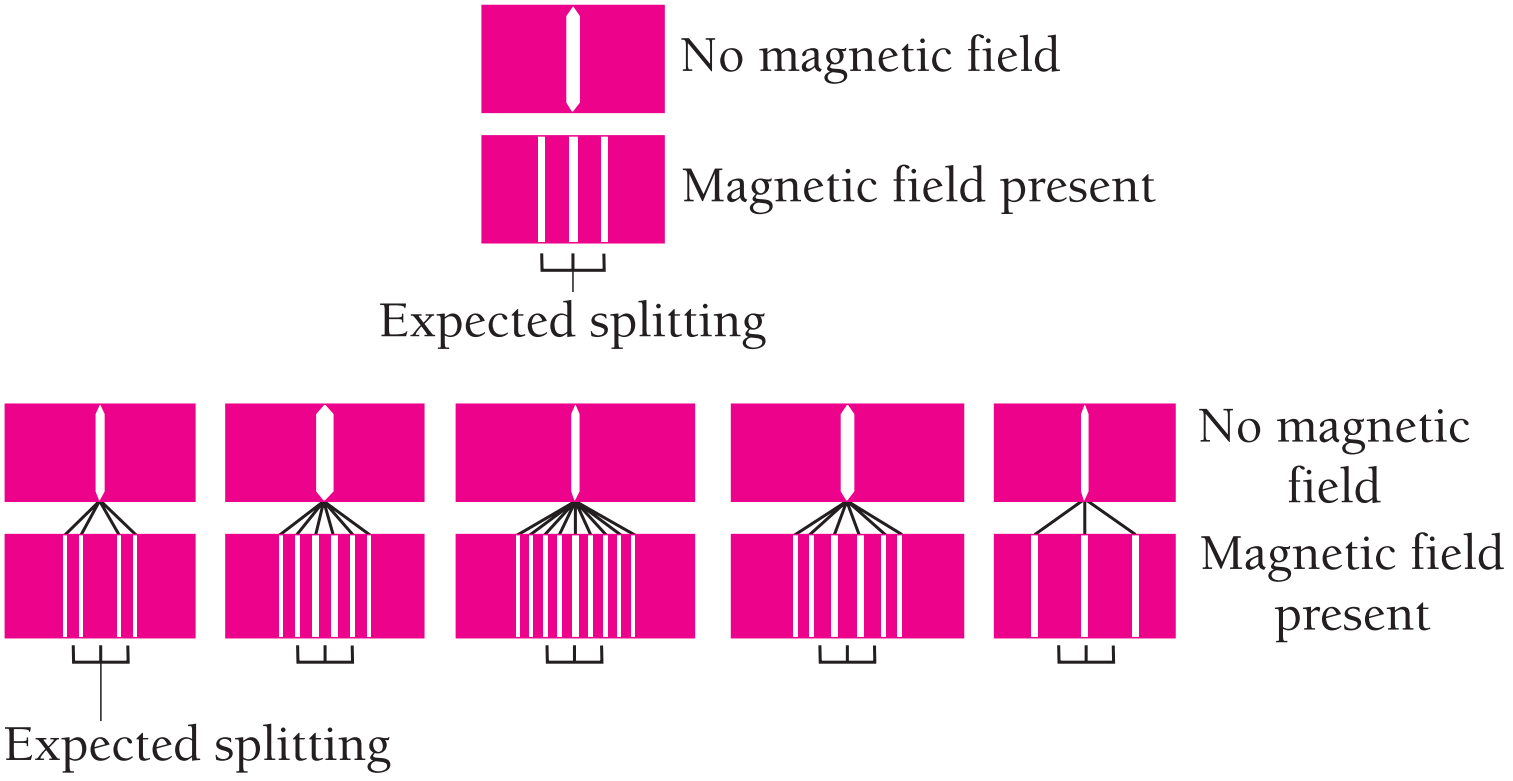
$$\nu_L = \frac{\omega_L}{2\pi} = \frac{e}{4\pi m} B(\mathrm{T}) = 14 B(\mathrm{T})\ \mathrm{GHz}$$

Zeeman effect



Anomalous Zeeman effect

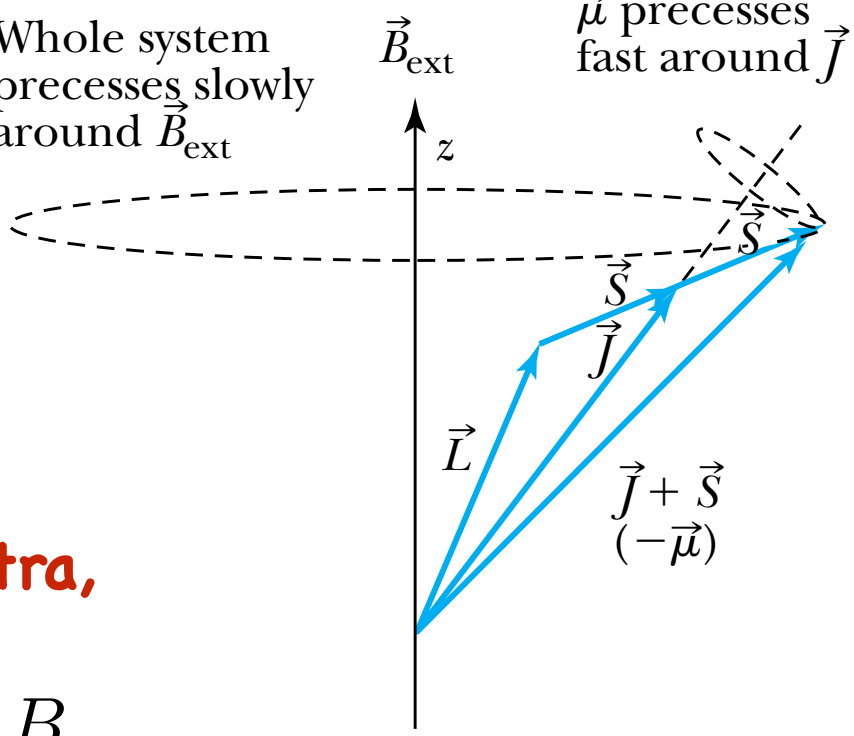
Soon after the discovery of this effect by Zeeman in 1896, it was found that often more than three closely spaced optical lines were observed.



4.7Anomalous Zeeman effect

We shall see that the anomalous effect depends on the effects of electron intrinsic spin.

$$\vec{\mu}_j = \vec{\mu}_l + \vec{\mu}_s,$$
$$= -g_j \mu_B \vec{J}/\hbar$$



The energy in magnetic field is,

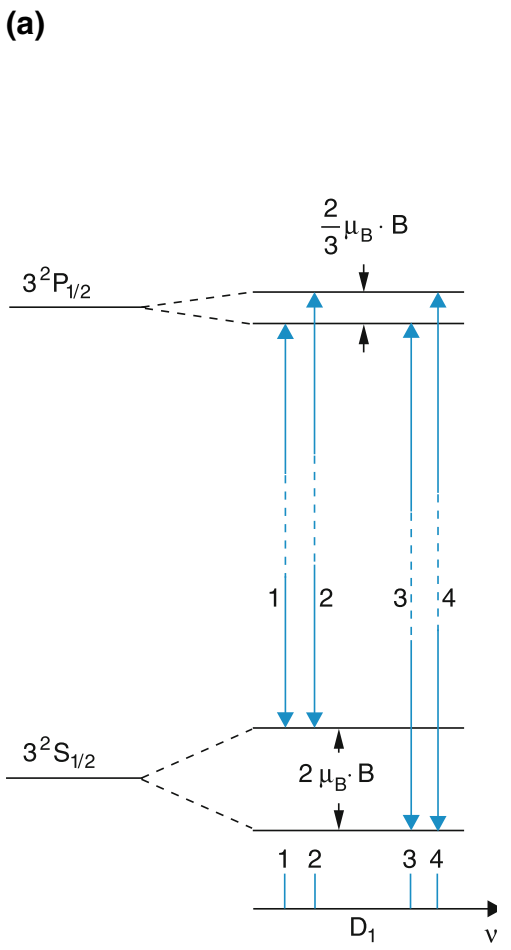
$$E_B = g_j m_j \mu_B B$$

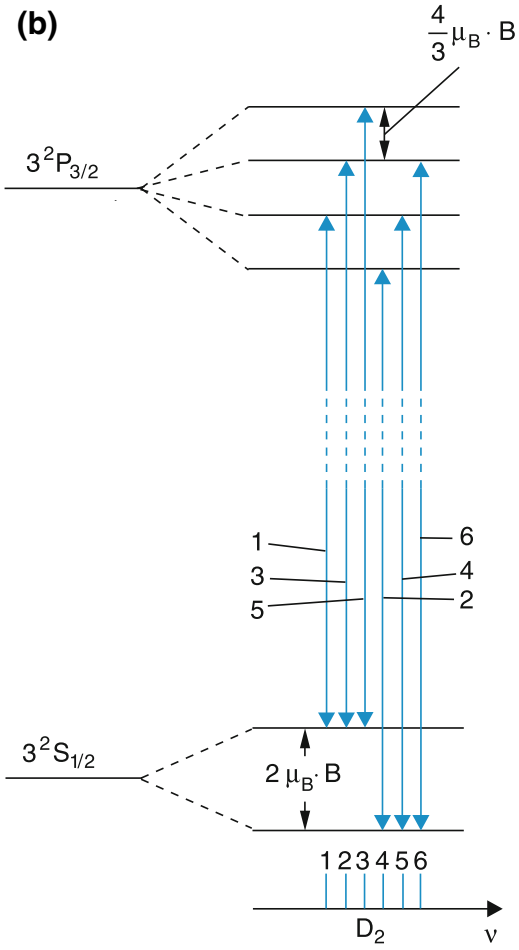
The frequency of splitting spectra,

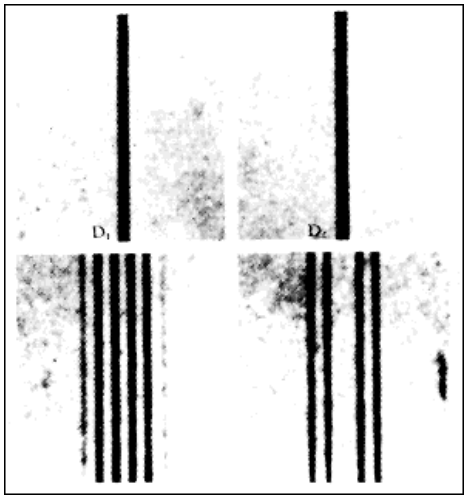
$$\nu = \nu_0 + (m_2 g_2 - m_1 g_1)\frac{e}{4\pi m} B$$

Anomalous Zeeman effect

The Zeeman pattern of the two D-lines in the sodium spectrum

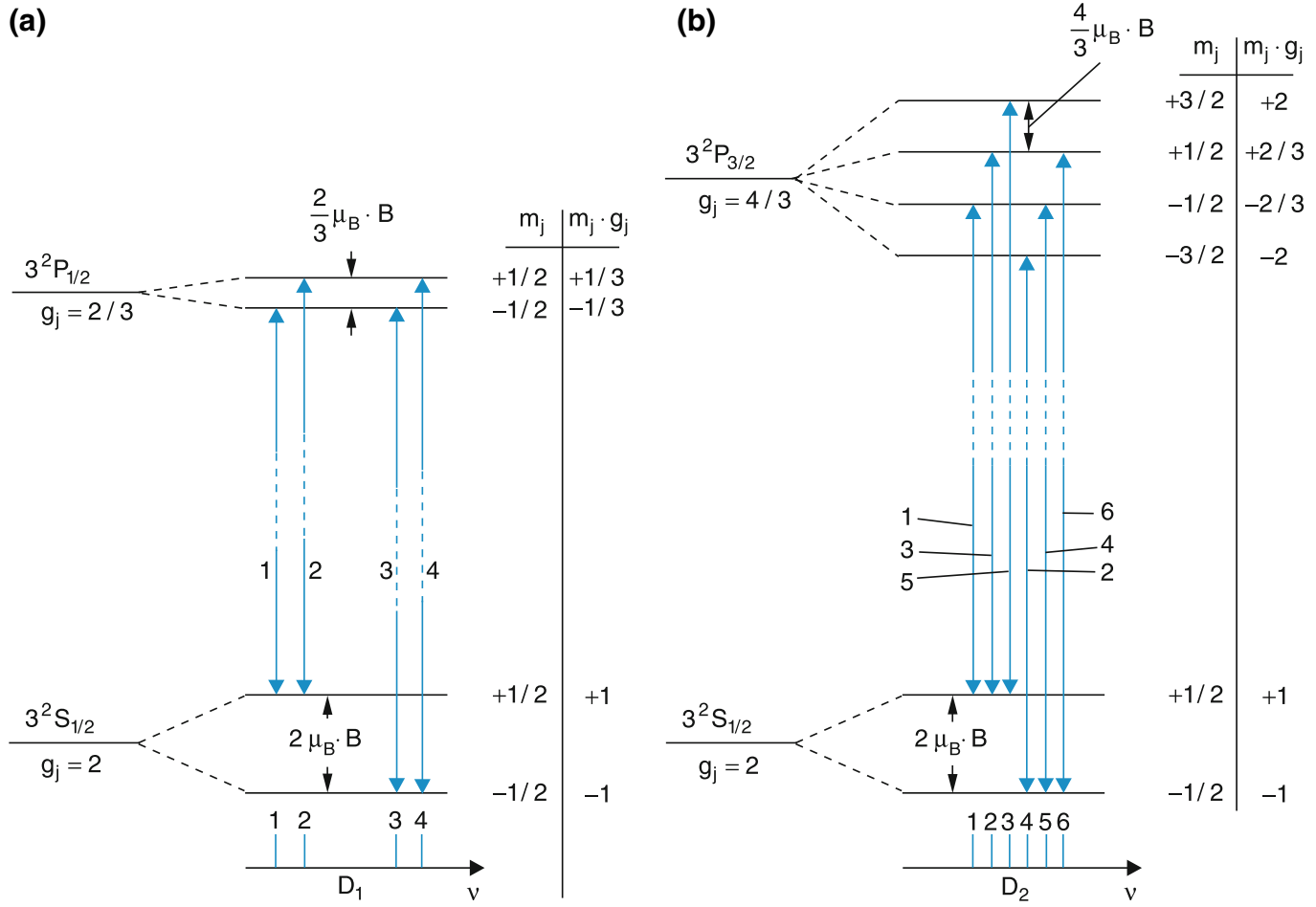




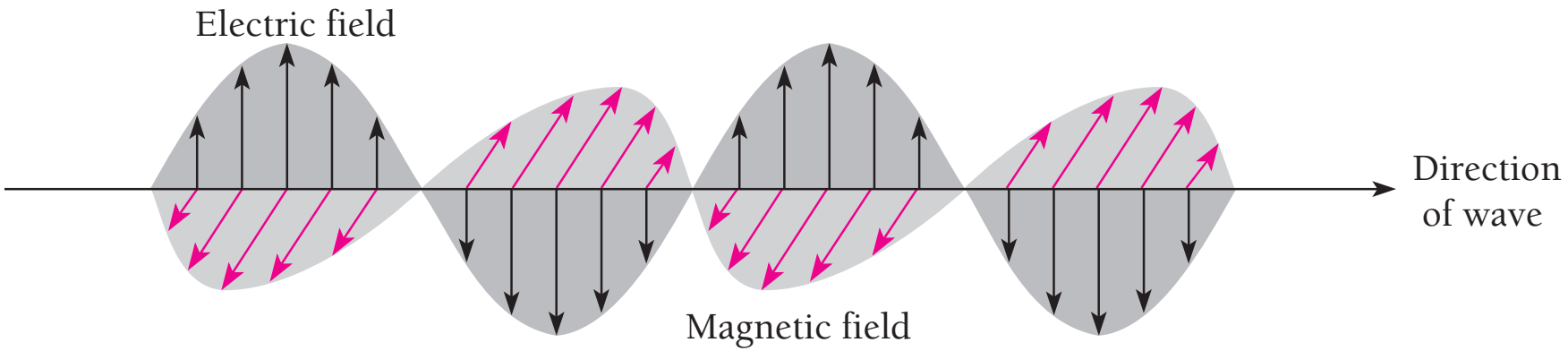


Anomalous Zeeman effect

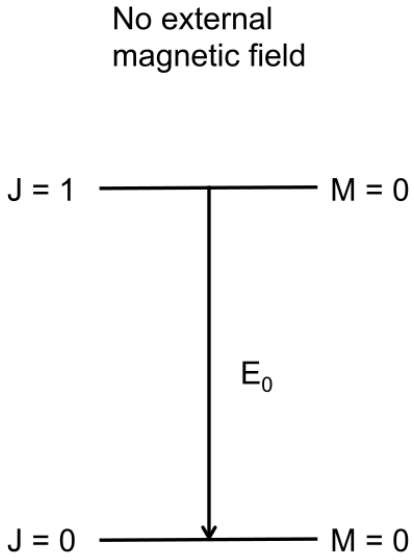
The Zeeman pattern of the two D-lines in the sodium spectrum

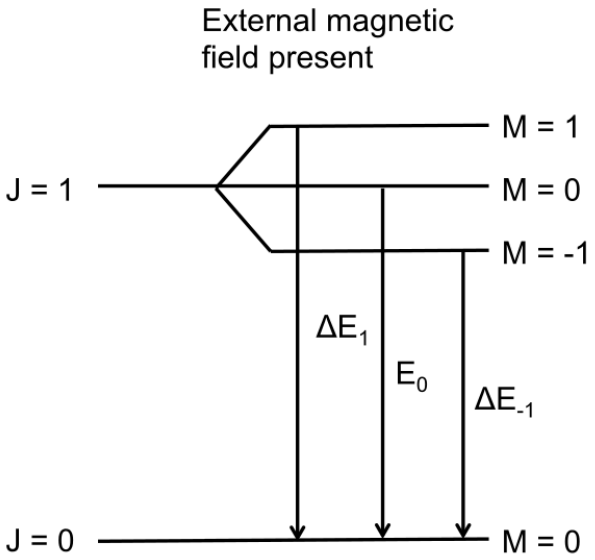


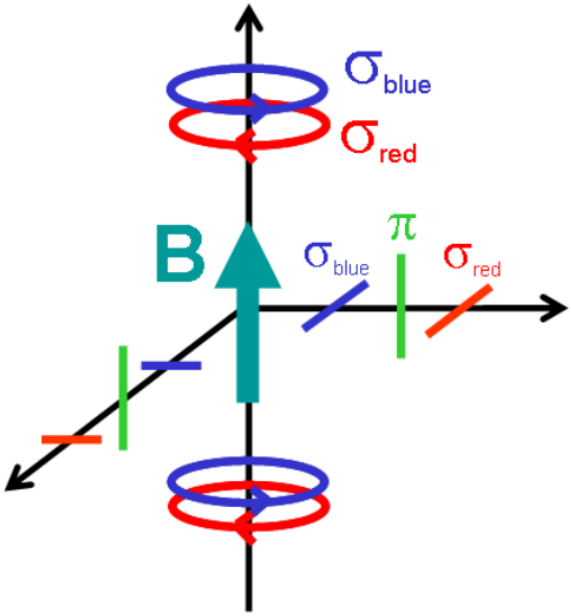
Polarization of light

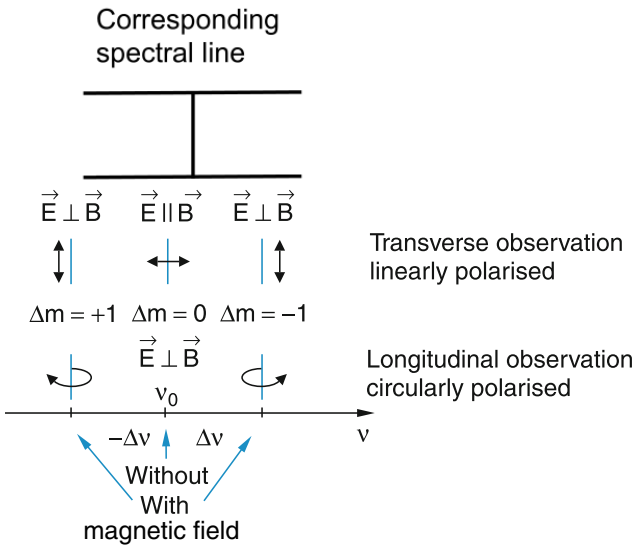


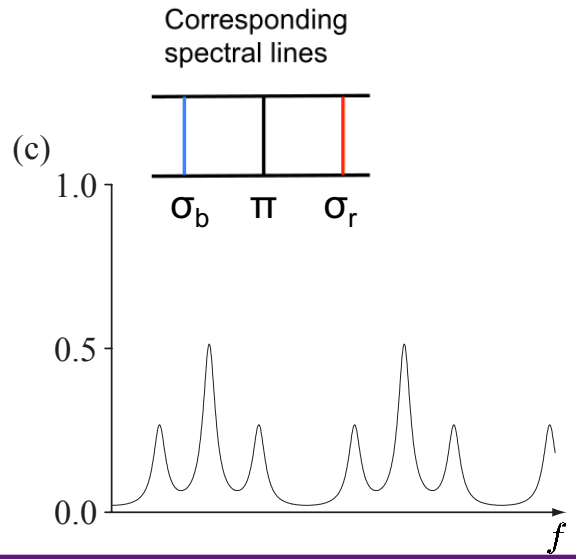
Polarization of Zeeman effect

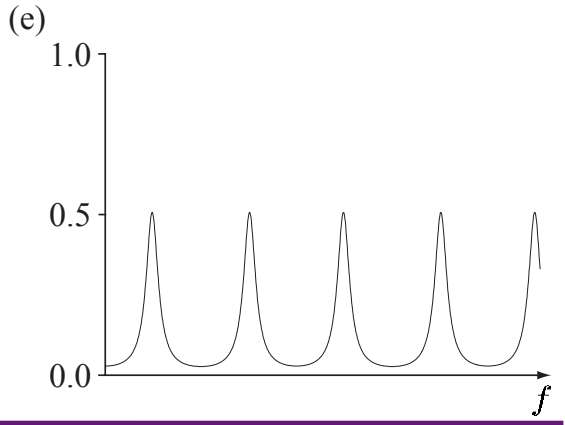






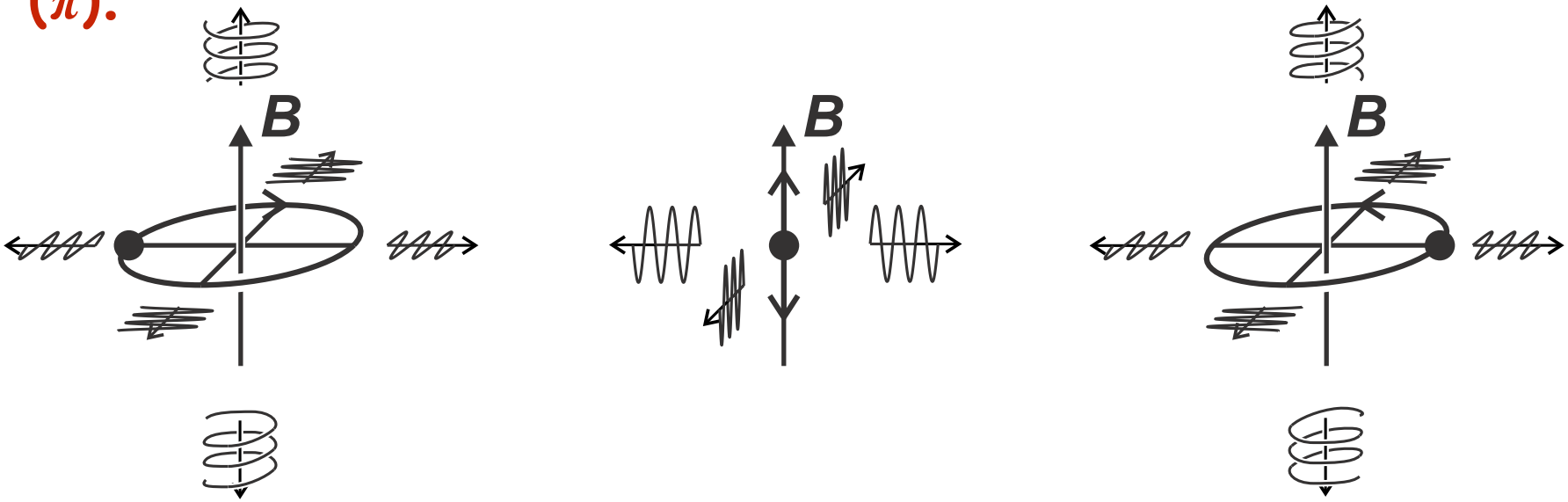






Polarization of Zeeman effect

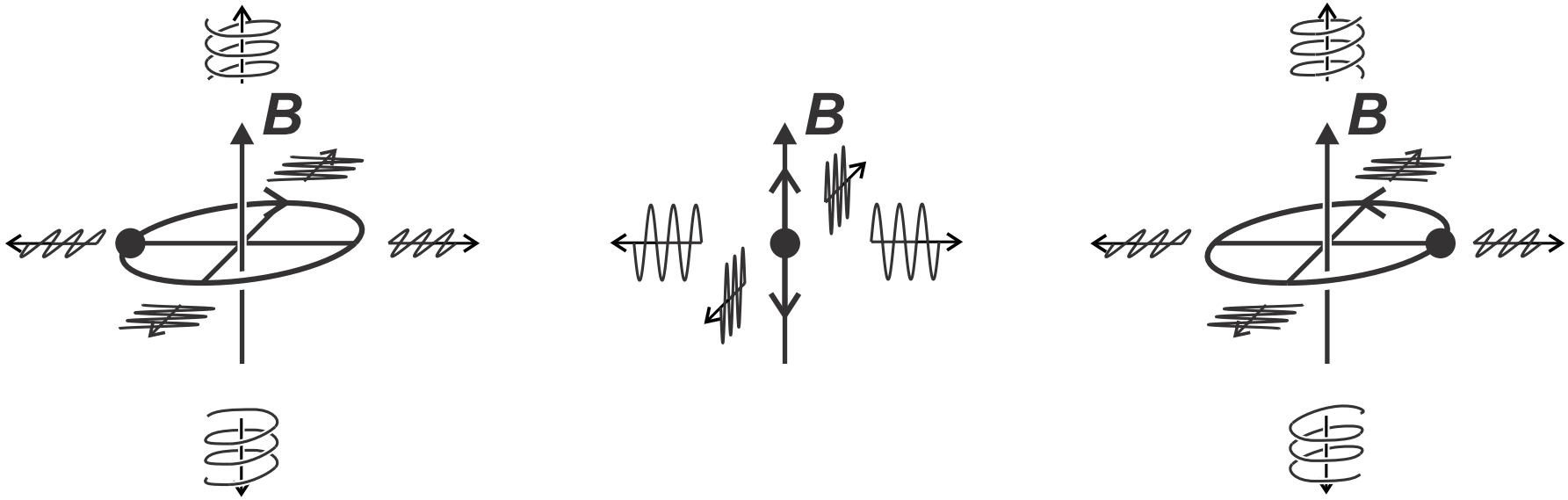
$\Delta M_J=0$ corresponds to an infinitesimal dipole oscillating parallel to the magnetic field. No quanta are emitted in the direction of the magnetic field. The light emitted perpendicular to the magnetic field is linearly polarized (π).



σ^- (ΔM_J = -1) π (ΔM_J = 0) σ^+ (ΔM_J = +1)

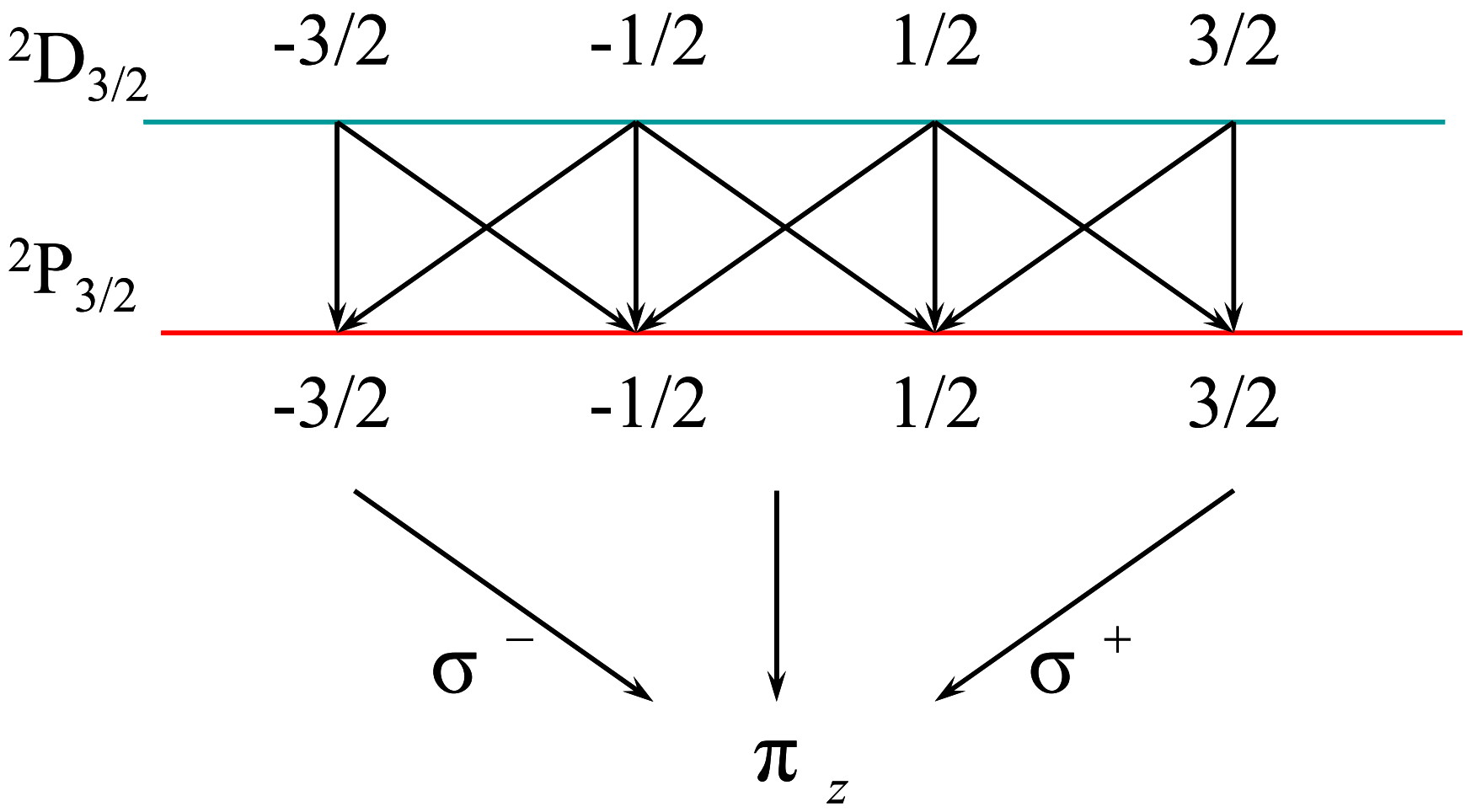
Polarization of Zeeman effect

$\Delta M_J=1$ corresponds to two parallel dipoles oscillating with a phase difference of 90^0 . The superposition of the two dipoles produces a circulating current (σ^+, σ^-).



σ^- (ΔM_J = -1) π (ΔM_J = 0) σ^+ (ΔM_J = +1)

Grotrian Diagram



原子物理学

4-2, 4-10

The Physics of Atoms and Quanta

12.1, 12.3, 12.5, 13.1, 13.2, 13.4

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The Physics of Atoms and Quanta (H. Haken, H. C. Wolf)

12.1 Calculate the precessional frequency of electrons and of protons [$I = 1/2$, magnetic moment $= (2.79/1836) \cdot \mu_B$] in the magnetic field of the earth ($\simeq 2 \cdot 10^{-5}$ tesla).

12.3 How large is the magnetic field generated by the electron in the ground state of a hydrogen atom, at the position of the proton if it would circulate according to Bohrs model on the shell $n = 1$?

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The Physics of Atoms and Quanta (H. Haken, H. C. Wolf)

12.5 Calculate the spin-orbit splitting of the states of the hydrogen atom with $n = 2$ and $n = 3$ using the relations

$$V_{l,s} = \frac{1}{\pi}\frac{Ze^2\mu_0}{8\pi m_0^2 r^3}(s \cdot l),$$

and

$$r^{-3} = \frac{Z^3}{a_0^3 n^3 l(l+\frac{1}{2})(l+1)}.$$

What are the values for a Rydberg state with $n = 30$ for the largest ($l = 1$) and the smallest ($l = 29$) splitting? a_0 is the radius of the innermost Bohr orbit.

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The Physics of Atoms and Quanta (H. Haken, H. C. Wolf)

13.1 What frequency is required to induce electron spin transitions from the parallel to the antiparallel configuration, or vice versa, if the magnetic filed is 10^{-1} tesla?

13.2 Why is the ${}^4D_{1/2}$ term not split in a magnetic field? Explain this in terms of the vector model.

The Physics of Atoms and Quanta (H. Haken, H. C. Wolf)

13.4 The spectral lines corresponding to the $3p \leftrightarrow 3s$ transition in sodium have the wavelengths $\lambda_2 = 5895.9$ Å and $\lambda_1 = 5889.6$ Å.

a) Calculate the magnetic field strength at which the lowest Zeeman level of the $^2P_{3/2}$ term would coincide with the highest level of the $^2P_{1/2}$ term, if the conditions for anomalous Zeeman effect were still fulfilled.

b) How large are the frequency differences between the outer two components of the D_1 line and of the D_2 line in a magnetic field of 1 tesla?